





Fundamentals of Additive Manufacturing

Groover M.P.: Chapter 29

until now $\Delta m = 0 \rightarrow$ $\begin{cases} \text{metal forming} \\ \text{metal casting} \\ \text{polymer shaping} \end{cases}$ Volume does NOT change

CNC machine $\leftarrow \Delta m < 0$ (material removing processes)

processes based on adding material $\leftarrow \Delta m > 0$ Additive manufacturing $\rightarrow$ 3D PRINTING
 $\downarrow$
technologies enable to shape the material as desired adding material



LEARNING TARGETS

2

{ IDEA OF WHAT AM IS? }

The student:

- knows what AM is;
- knows the AM process chain;
- knows the AM process categories.

Additive
Technologies,
what are the processes,

We can shape
many different
materials with
known technologies
↑
metals, ceramics, polymers...

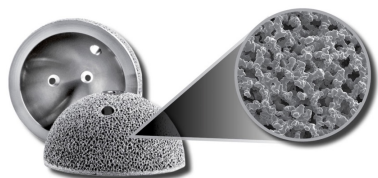
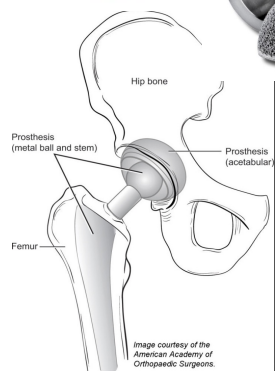
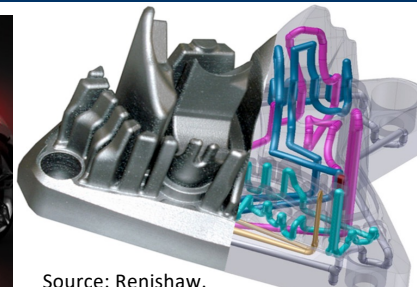




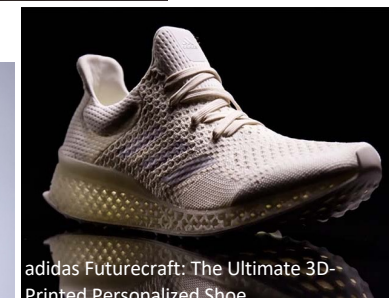
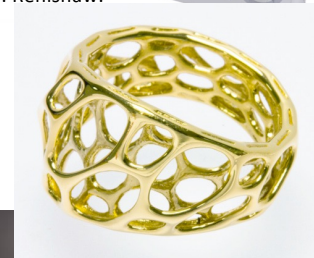
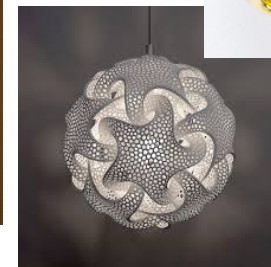
WHAT DO THEY HAVE IN COMMON?

↪ type of possible products...

3



<https://www.eos.info/dental>



COMPLEX GEOMETRY!
PRODUCT

$\Delta m > 0 \rightarrow$ add material
 $\Delta m < 0$

from $\Delta m = 0 \rightarrow \Delta m > 0$

selectively adding material $\equiv$ more complex

complex object $\Rightarrow$ capability of printing the material...

FASTER process $\Rightarrow$ shape channels

Complex STRUCTURE

shape able to fit well the body (prothesis) $\rightarrow$ Lots of possibilities
 $\downarrow$
personalized objects (shoes, prothesis, ecc...)

speaking of
{ INJECTION } $\rightarrow$ channels inside the mold to control Temp
{ HOLDING } (straight Holes...) impossible to be close to real object

If able to shape channel to follow complex geometry $\rightarrow$ capability to control Temp. will be higher $\rightarrow$ FASTER, EFFECTIVE
 $\Downarrow$

but to shape complex channel is difficult! we may add material by material (layer by layer)

leave space $\rightarrow$ have complex structure internal

complex shapes / personalized objects like SHOES (by scanning the foot)



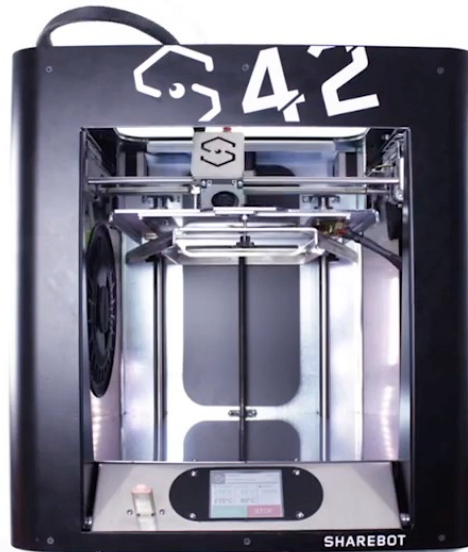
WHAT IS ADDITIVE MANUFACTURING?

DEFINITION STANDARD

Additive manufacturing: “the process of joining materials to make parts from 3D model data, usually layer upon layer, as opposed to subtractive and formative manufacturing methodologies.”
(additional way to shape material..)

VIDEO EXAMPLE

3D PRINTER
(adding polymer material, melt and shaped cooling down)



adding plastic polymers (cold material)
selective add material → cool down
↓
solidification



ADDITIVE MANUFACTURING – DESIGN FREEDOM

speaking about geometry COMPLEXITY



e.g. bamboo structure

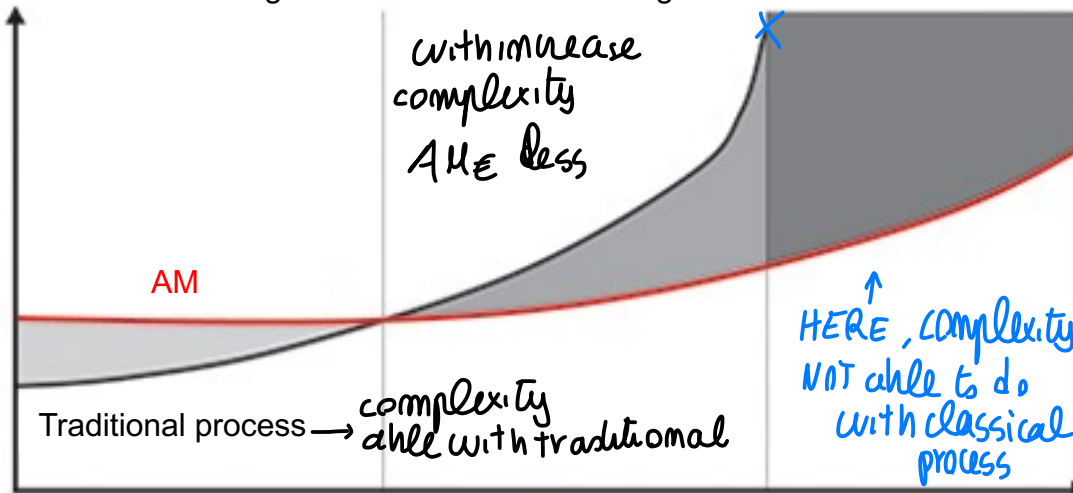
- Lightweight
- Less material
- Improved mechanical properties
- More durable

Traditional process
cost advantage

AM process
cost advantage

AM design advantage

Manufacturing
cost per part



With increasing complexity, some geometry can't be produced by traditional processes

comparing Additive manuf.
respect traditional processes
depending on the cost

(industry ask if some product is better to do
with classical manufacturing or AM)



ADDITIVE MANUFACTURING – DESIGN FREEDOM

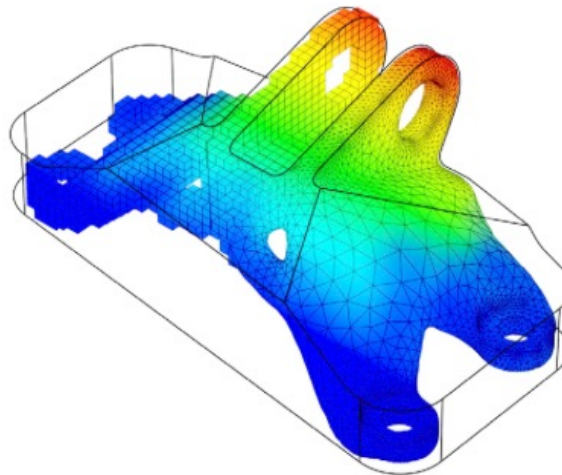
EXAMPLE

↓ COMPLEX object → functional geometry, sometimes is better to reduce the weight



GE's original design

The original design of the bracket used in the GE Aircraft challenge. The re-designed bracket was required to fit within this volume and meet load requirements while reducing the overall weight of the part.



Cloudmesh optimization

The topology optimized part design generated by Frustum's software.

difficult part to make, possible with AM



Final DMP part

The GE Aircraft engine bracket redesigned by Frustum and manufactured by Quickparts on a 3D Systems ProX DMP 320 printer. The new part passed all the load condition requirements specified by the GE challenge and stayed within the same footprint while reducing weight by a staggering 70 percent.

in lot cases I need light weight design

⇒ reduction of weight, with same requirements



ADDITIVE MANUFACTURING – DESIGN FREEDOM

Michelin has revealed its Vision concept tyre at the 2017 Movin' On conference in Montreal, Canada.

The new visionary Michelin concept tyre is a 3D-printed airless, biodegradable tyre, connected, rechargeable, customizable and organic as well.

(June 2017)

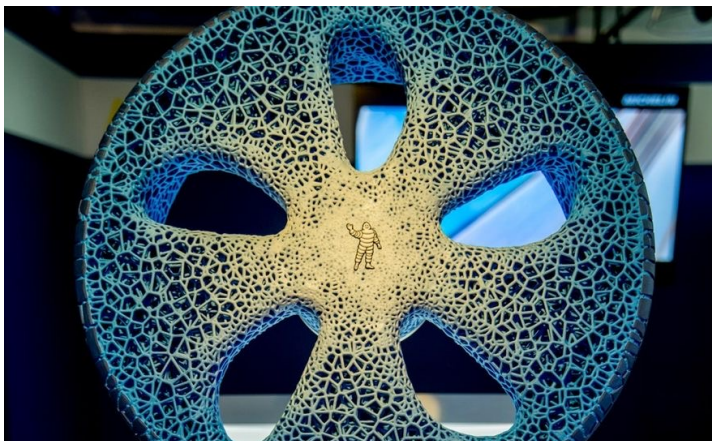
rethinking
the wheel

NOT
possible
to produce
without
AM



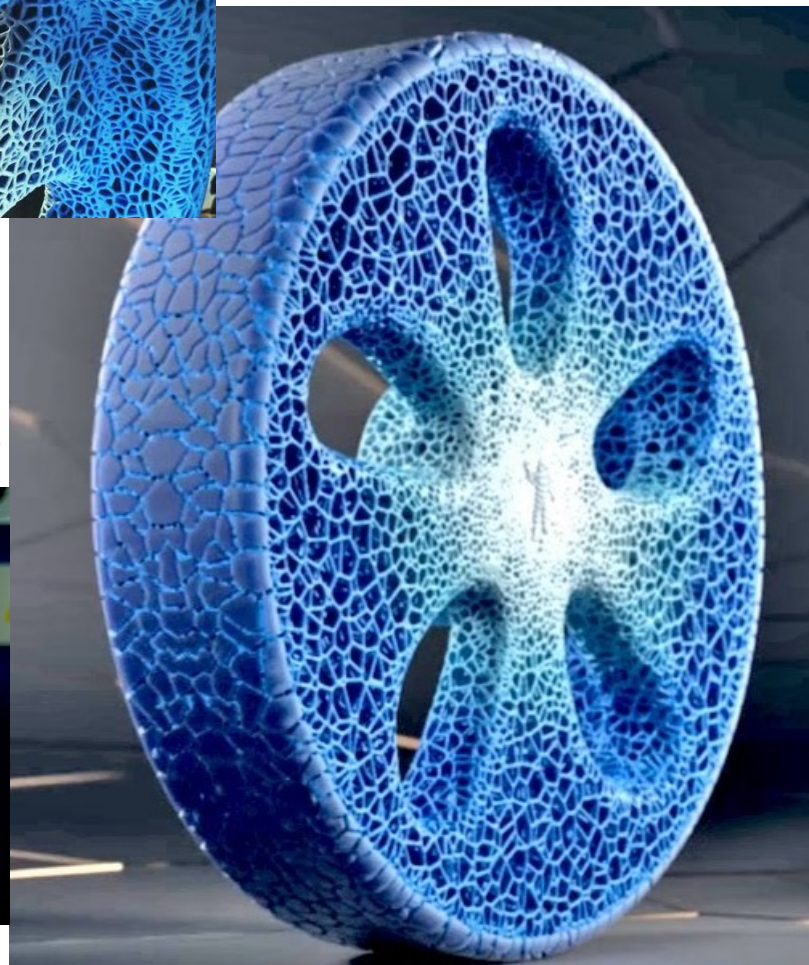
FUTURE Michelin
Wheel → STRONG
complex

last layer Reshapeable
by removing or reprinting



HIGH
COMPLEXITY

polymeric
↑
JUST of an unique part



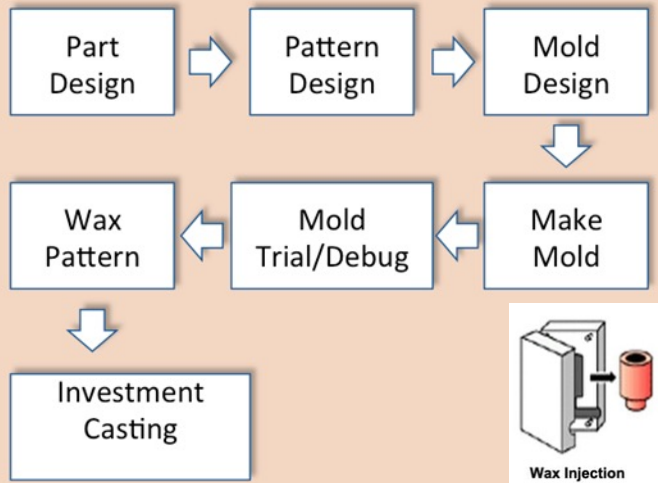


ADDITIVE MANUFACTURING – DESIGN FREEDOM

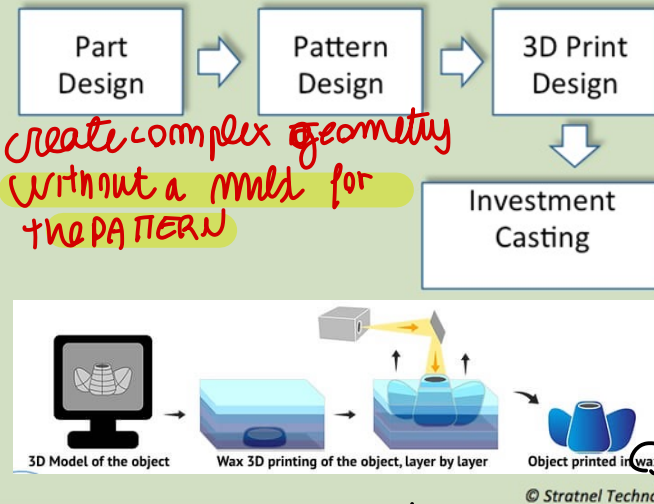
Rapid Tooling

↓ AM can be used to create part = device for classical manufacturing processes

Investment Casting – Traditional Flow



Investment Casting – 3D Printing Flow

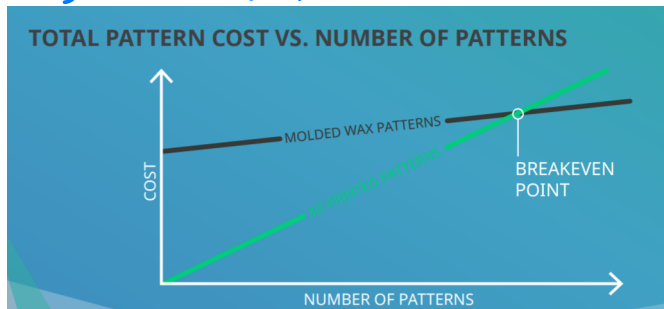


Why don't we produce the pattern with AM

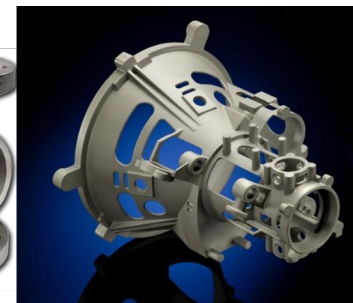
No open mold problem
additive manufacture

sometimes is better to have the mold.

Directly from PATTERN DESIGN to make it



Source: <https://www.3dsystems.com>



for high complexity of pattern, impossible with AM

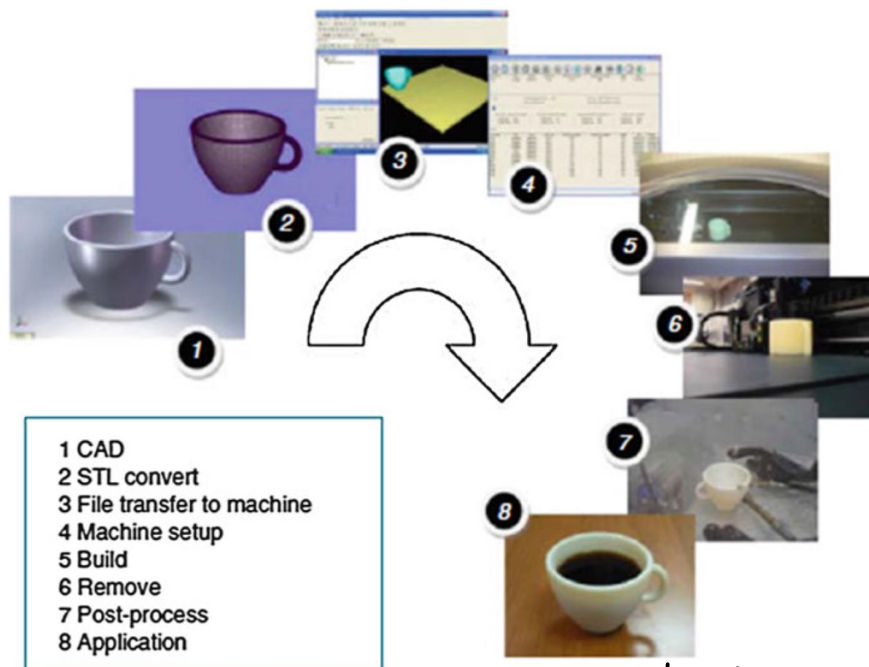


AM PROCESS CHAIN

PROCESSES in which we have
different steps

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The process chain involved in additive manufacturing technologies is characterised by **rapid, flexible and direct fabrication from 3D CAD data**. Intermediate stages, such as tool manufacturing, are **not necessary**.



from Part Design *theoretically* directly to solid model used on machine process, AM (growing the part inside machine)

There are basically two different categories:

single-step processes: parts are fabricated in a single operation where the **material properties** of the intended product are achieved; **NET-SHAPED**

multi-step processes: parts are fabricated in two or more operations where the **first** typically provides the **basic geometric shape** within the additive manufacturing machine and subsequent **post-processing** enhances the part's material properties (e.g. using infiltration, sintering, or other thermal or chemical modification).

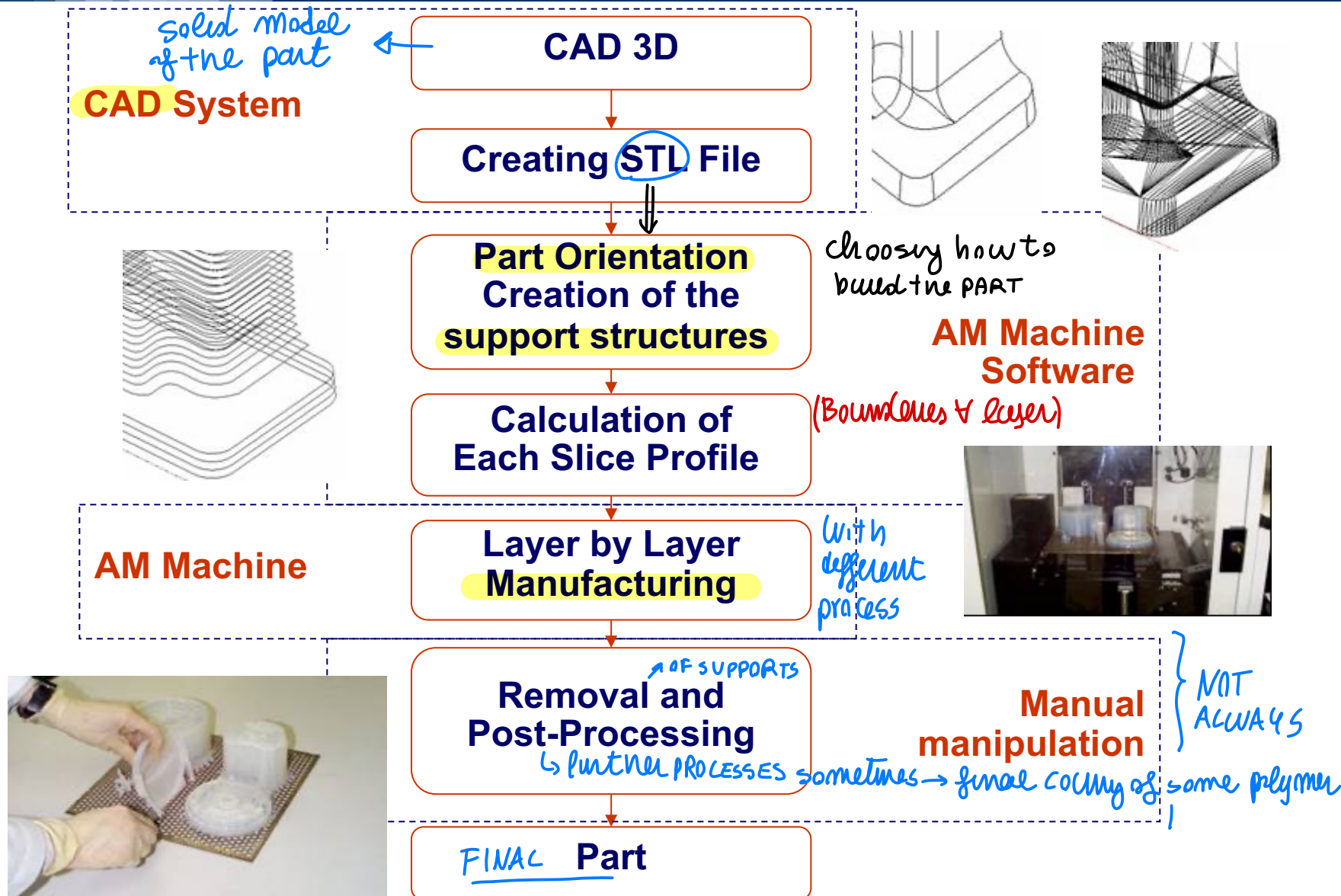
POST processing required for final part



AM PROCESS CHAIN

represents the boundaries
of my product

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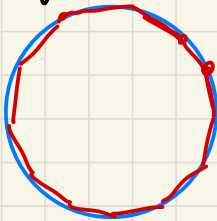
STL file

representation of the solid boundaries

set of connected points → approximate the boundaries of the object with a planar surface

smaller = more accurate → simple way to represent INTERNAL/EXTERNAL boundaries

like, if given a circle



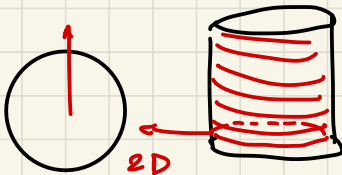
represent it as set of segment points and connection among points

} better approx depending on how points are displaced

STL file as input of AM machine

→ cause, once defined the geometry I have to deal with the AM process, so define the growing direction

Adding layer by layer from reference plane



boundaries of CROSS-SECTION

Set of parallel

lines → do CROSS SECTION of plane

you can have complex internal feature

→ add material only where needed

↓
add material With small layer but with → STAIR CASE effect during AM

to have a good shape → layer thickness is important

small layer $\rightarrow$ more accurate $\rightarrow$ more time needed!
(more segment)

from SLT file $\Rightarrow$ you choose **direction of GROWING**

path orientation in machining

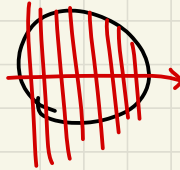


grow layer vertically
 $\Rightarrow$ BETTER than
horizontally



better than

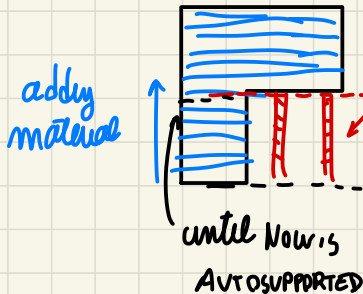
optimize
final quality
by good orientation



that direction of grow.
HIGH STAIRCASE
effect

+ sometimes we need **SUPPORT structure**

in fact $\Rightarrow$ creating an "L" shape



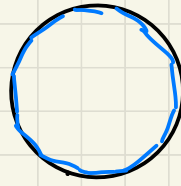
you need
support on that
point

additional material

for support
 $\downarrow$
additional
element to
support
the part

approximating by a representation

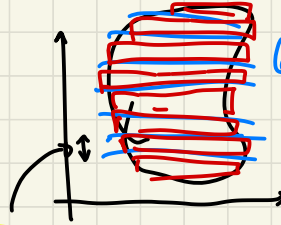
↳ using a series of lines



adding material layer by

a layer I need CROSS-SECTION

[deposit the material]



layer by layer!

shape piece by piece
the material

smaller layer lead to
a better quality (more segmentation)

CYLINDER

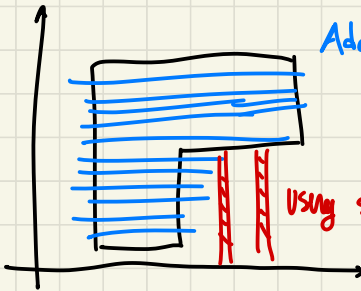


NOT segment
in this
direction



better layerization

IMPORTANT technique direction
on which put the layer!



Adding material →

using supports to NOT let
fall the product
during the process



AM PROCESS CATEGORIES

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Very different kind of processes

According to ISO and ASTM, the AM processes may be grouped in **7 basic categories**:

1. Vat photopolymerization
2. Material jetting
3. Binder jetting
4. Powder bed fusion
5. Material extrusion
6. Directed energy deposition
7. Sheet lamination

(very spread possibilities)

↙ group different

Technologies in 7 categories

*↳ OVERVIEW
of these techniques
types of process,
PRO/CONS, what material*



AM PROCESS CATEGORIES AND FEEDSTOCK MATERIALS

Valid
schema,
you can see
Material
that can
be used on AM
different
application

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.

one tech
to many
material or
one to one

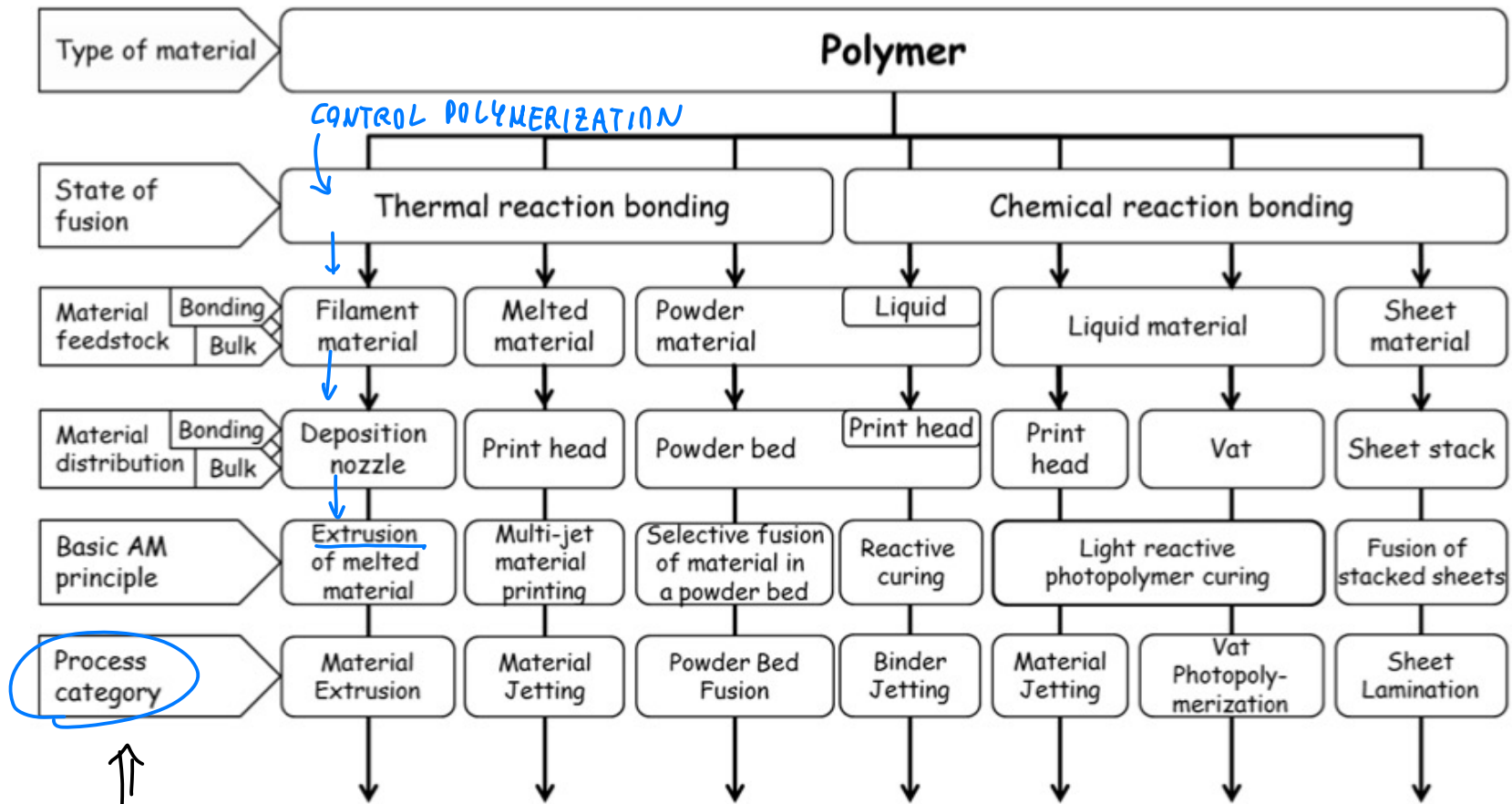
← just a
couple

← lot of
usage
process

↓
thermoplas
is easier
to shape



AM PROCESS CATEGORIES



Looking on POLYMER, you can classify the processes respect different behaviours



AM PROCESS CATEGORIES

↑ FIRST PROCESS that can be classified as AM

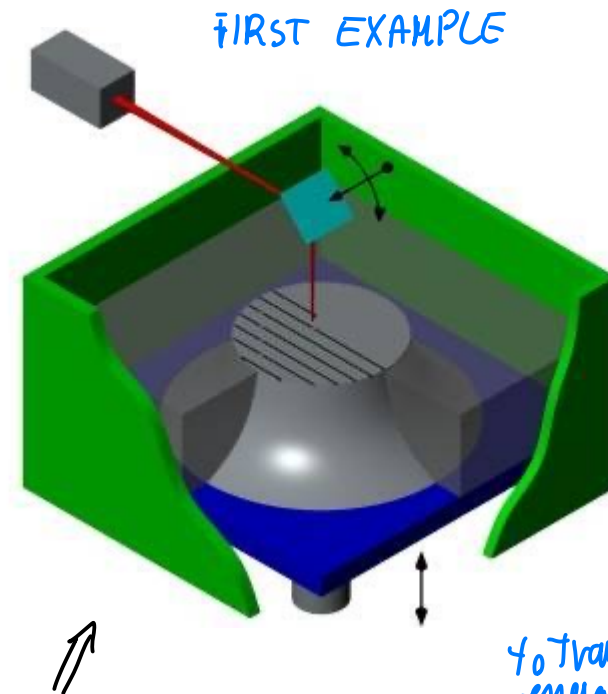
↑ STEREO LITHOGRAPHY

Vat photopolymerization: an additive manufacturing process in which liquid photopolymer in a vat is selectively cured by light-activated polymerization.
Liquid → energy → solidification

COVERING
↓

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.



FIRST EXAMPLE

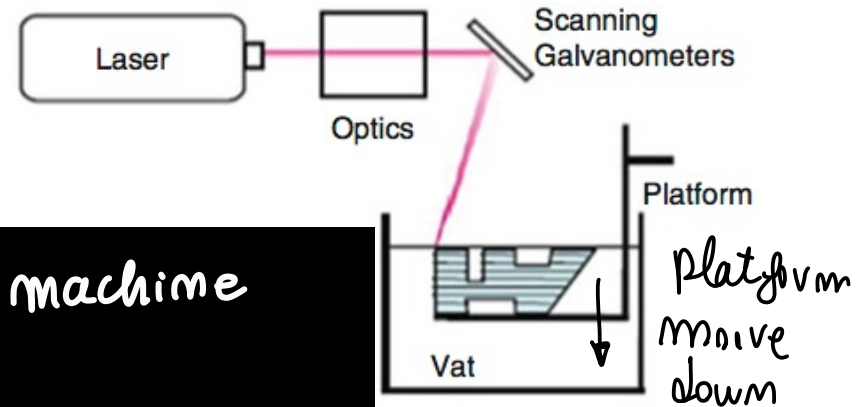
↑ to transfer energy to material

First AM system ever: SLA1 3D System 1987 (Charles (Chuck) Hull)

VAT with liquid polymer (pre-polymer) → selectively COVERING the MATERIAL using UV Light activate CURING



Vector Scan VP Machines: Stereolithography (SL)



build the part one layer a time by a machine
used in a lot of sectors → like [↑] AUTOMOTIVE
MEDICAL
Etc..

⇓
liquid introduced @ the surface, starting the forging
Addition {polymerization} ⇒ UV LASER to transform
and scan the surface
(FAST CONTROL) ← by MIRRORS controlled to direct
properly the light

let the liquid
solidify upside the
previous layer
↓
model the mold material
COMPLEX SHAPE

→ build the
structure

<https://youtu.be/NM55ct5Kwil>

Decomposition and control of each Layer...
Multiple CONTROL PROBLEMS { temperature, movement, melting, etc }

SL: rapid process to create parts from 3D parts

3D PRINTING of AM

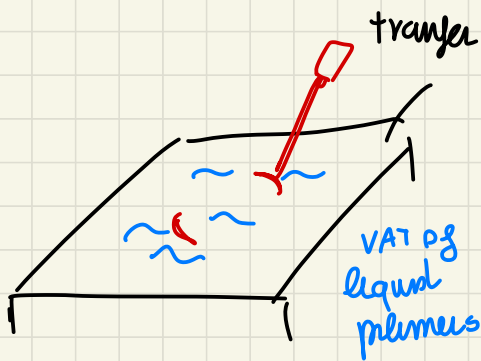
build one layer @ time $\rightarrow$ each part construct by LASER

Laser trace and form each layer inside liquid polymer

then part is COURED on another part

$\downarrow$

than handly WORKED



transfer energy to the material
 $\downarrow$
photo polymerization
process $\rightarrow$ chain reaction,
by photons interaction
by eliminating the surface

UV Light
UV Laser } LIGHT $\rightarrow$ UV Ray

+ surface scan by
MIRRORS

Galvanometric

MIRRORS $\rightarrow$ solidify
layer by layer

control position and direct the light

V layer CROSS-SECTION

create also the support out of polymer

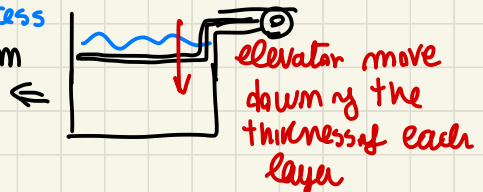
$\uparrow$ of same material of object

You may
minimize support
structure

$\rightarrow$ difficult to separate

during the process

go down of some μm



very thin layer of solid & passage

polymer solidify on the layer → model top plate

move down by layer dimension

Using blade to deposit material

↑
recording plate motion deposit material

layer by layer → build solid part structure

lot of element created is the support for complex shape

LOT OF CONTROLLING → blade, motion of elevator / Temperature, Penetration of melt, laser motion...



VAT PHOTOPOLYMERIZATION

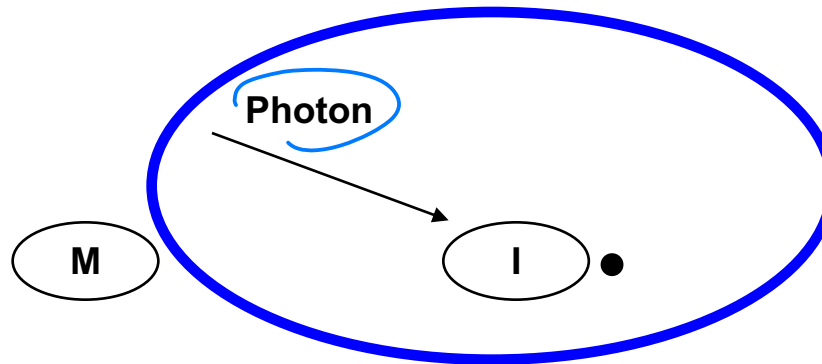
Photo-polymerization process

photon Interact with
Initiation → create a

Free radical
formation

Interact with
M_n-polymers
create a link... chain
reaction!

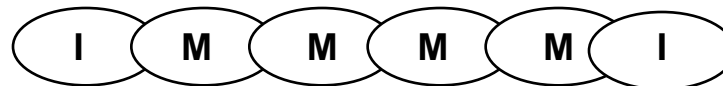
Initiation



Propagation



Termination



PHOTON starts chain reaction → why you don't have energy on all vat?

- Two main kind: you have possible ways that this polymerization ends → (NOT all vat solid)
1. Free-radical photo-polymerization (acrylate)
 2. Cationic photo-polymerization (epoxy and vinyl ether)



VAT PHOTOPOLYMERIZATION

Polymerization terminates from one of three causes, recombination, disproportionation, or occlusion.

- ↓ from a free radical → two macromolecules JOIN
- **Recombination** occurs when two polymer chains merge by joining two radicals. → JOIN two radicals, any one able to create new
 - **Disproportionation** involves essentially the cancelation of one radical by another, without joining. inhibit the reaction ↑
chemical ELEMENT interacts with free radic. , by exchanging atoms → chemical reaction
 - **Occlusion** occurs when free radicals become “trapped” within a solidified polymer, meaning that reaction sites remain available, but are prevented from reacting with other monomers or polymers by the limited mobility within the polymer network.

free radical inside a solid lock → potential way to add material → but blocked

chemical / physical situation that STOP THE PROCESS → continue SCANTO give energy for continuously solidify



VAT PHOTOPOLYMERIZATION

process applied on..

UV curable photopolymers are thermoset polymers:

2 types of resin

1. Acrylate-based resins

both
they
have PRO & CON

- ☹️ High reactivity (photospeed) → FAST process
- ☹️ 5-20% shrinkage → inaccuracy (curl distortion)
- ☹️ Susceptible to oxygen inhibition (easy STOP)

(High photospeed → FAST process)

fast
curing

!!!

complex parts..

difficult to shape, shrinkage
problem... difficult
to compensate

2. Epoxy resins

- ☹️ Low photospeed → more time to solidify (MORE PROCESS TIME)
- 😊 1-2% shrinkage more accurate than ACRYLATE
- 😊 Not susceptible to oxygen inhibition → less stop problem
- Brittleness

3. Commercially available resins: epoxides with some acrylate content. They form a polymer blend rather than a copolymer: monomers undergo independent polymerization.

↑
mixed resins

BLEND with
both benefits

↑
material used on that Machine you have
CO-POLYMER, huge macro molecule of ACRYLATE
+ EPOXY

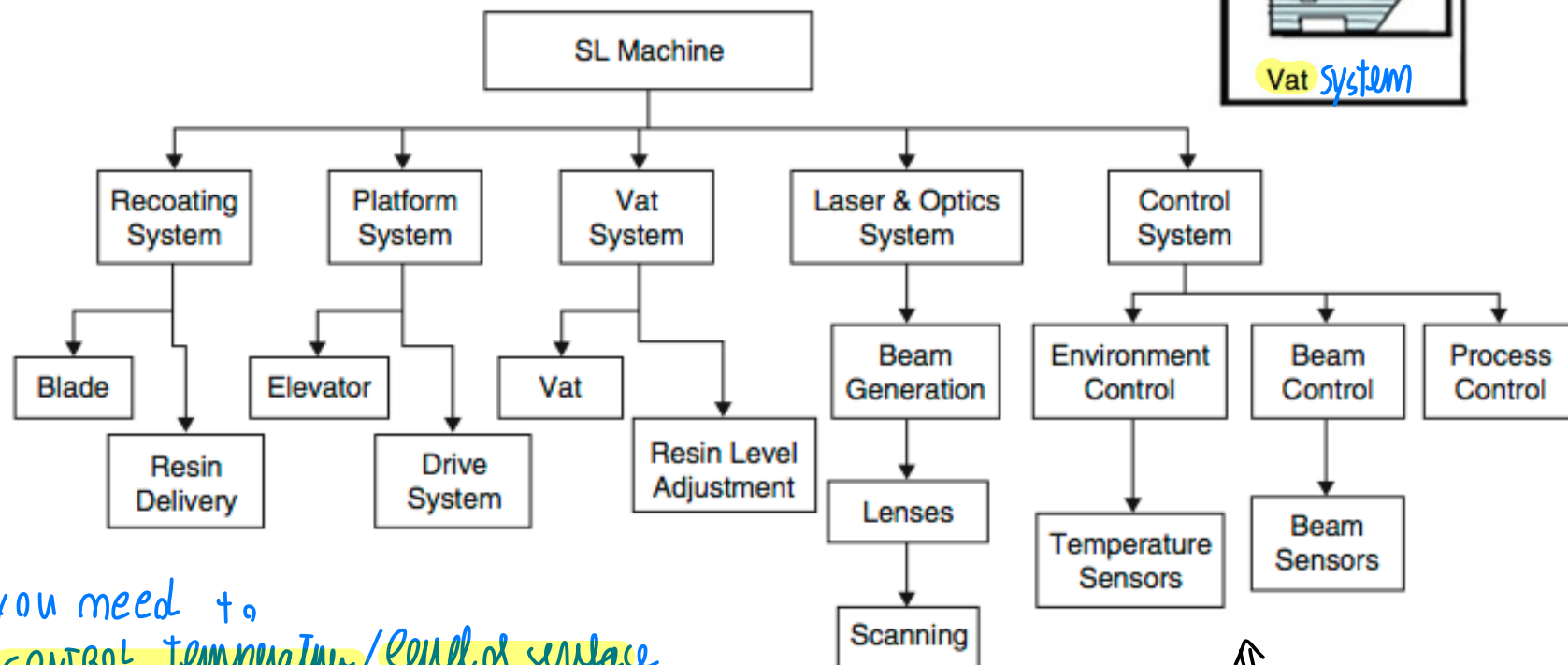
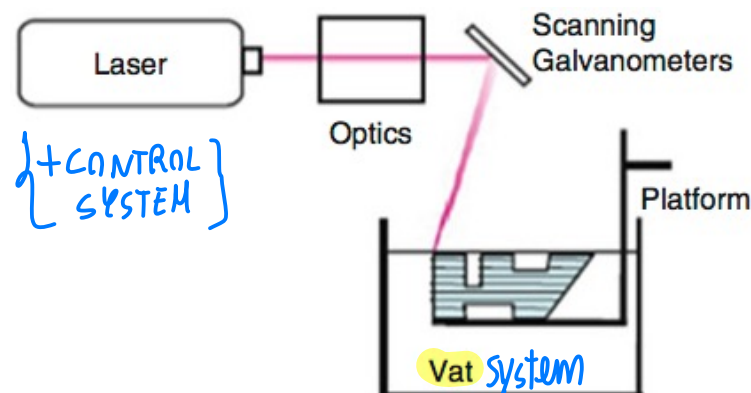


ELEMENTS OF THAT MACHINE ... VAT PHOTOPOLYMERIZATION

lot of aspect to CONTROL!

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Vector Scan VP Machines: Stereolithography (SL)



you need to
CONTROL temperature / level of surface
in the VAT

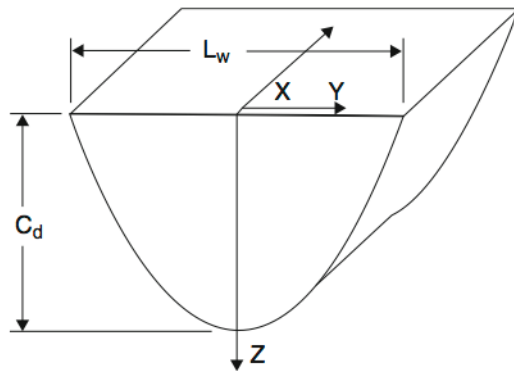
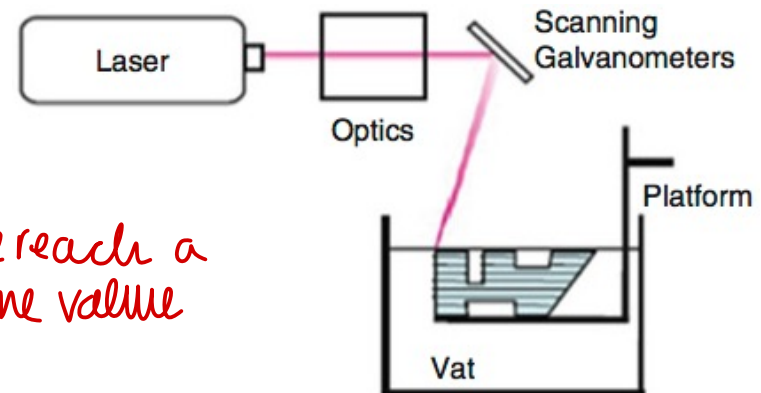
level of PLATFORM (linear motion), control optics generation of beam / scanning etc...



VAT PHOTOPOLYMERIZATION

Scan Patterns

INTERACTION
of the laser with
material
(transferring unit)

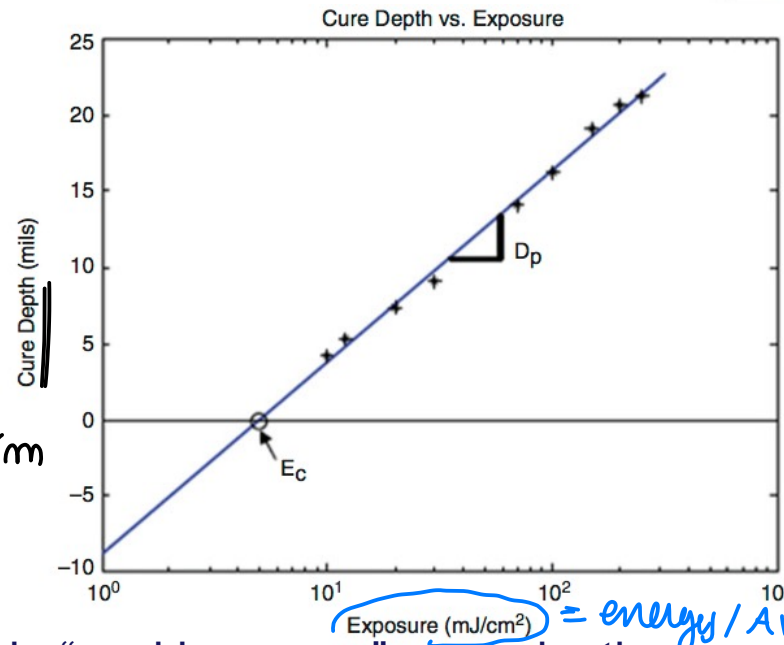


Cured line showing
parabolic shape

C_d : cure depth,

L_w : line width. how deep I'm
solidifying the
material

CURING if we reach a
certain exposure value



Exposure
||
Energy/Unit Area

solidification
Depth
↳

Resin "working curve": cure depth vs. exposure
(1 mil = 0,001 in. = 0,25 mm)



VAT PHOTOPOLYMERIZATION

enables to create boundaries...
then immer part solidly thanks
to stop criteria, INTERNAL SOLID → @ end I've some polymer melt
21
Inside

Scan Patterns

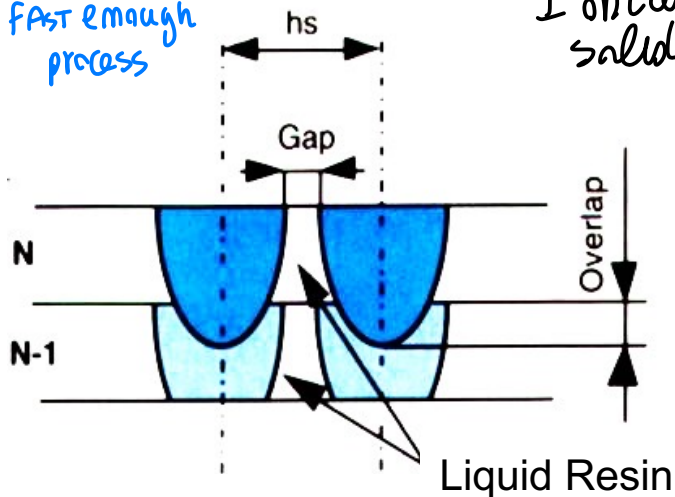
BHO

→ different scan strategy

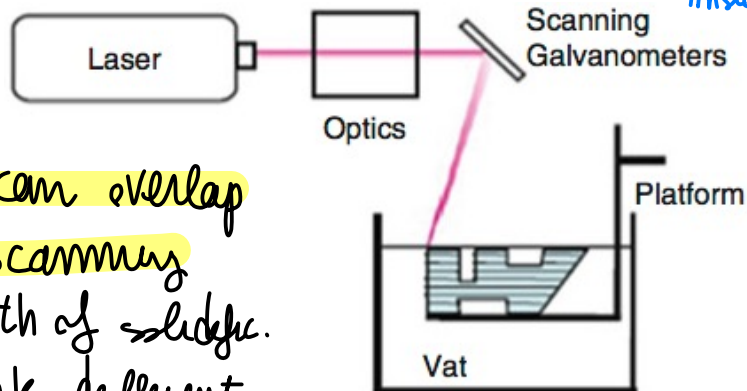


@ end
↓
final full
CURING
POST-PROCESS
FAST enough
process

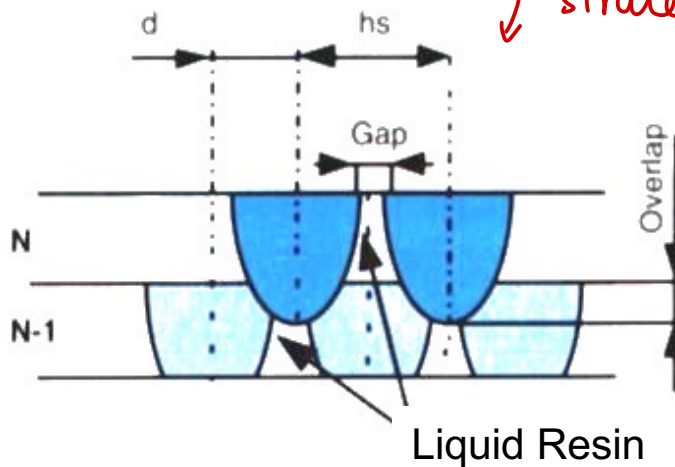
IF I scan the
surface completely
I obtain a
solid surface



We can overlap
2 scanning
(depth of solidification
to link different
layers) → OVERLAP (Gap between layers)



different
↓
strategies



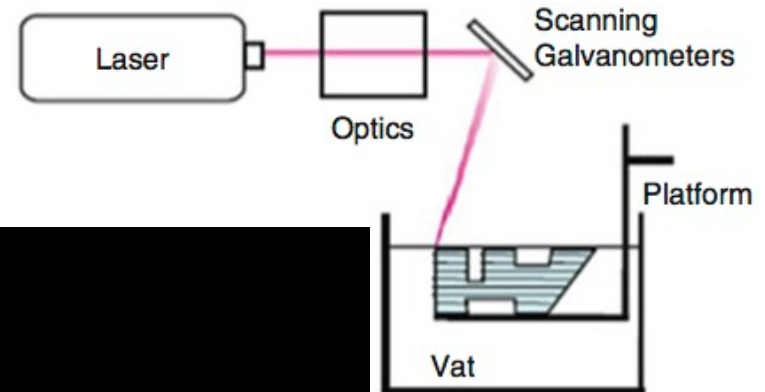
Solidify
different
portion



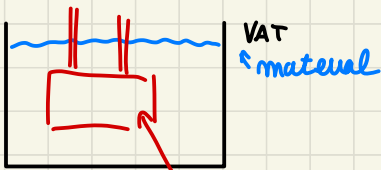
Vector Scan VP Machines: Stereolithography (SL)

@ end → further polymer through usv

Dimensions
of objects in SL → BIG



<https://youtu.be/V7PeN9F4Dns>



full solid stay inside
polymer melt, high
temperature!
↓

this can create further non
CONTROLLED chemical reaction
changing dimension

← you can try to avoid keeping the material inside
the VAT



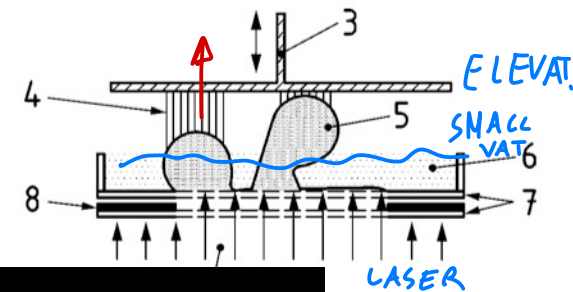
VAT PHOTOPOLYMERIZATION

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applied on
SMALL OBJECTS!
well done

Vector Scan VP Machines: Stereolithography (SL)

MOVING the
object OUTSIDE
the LIQUID,
SO NO MORE
Reaction



JUST
USING
AIR,
push liquid
Below
the solid
and SCAN
surface
from
Below

VAT on SL have high Temp liquid

↓
sometimes it deform the solid shape

↓ How to NOT have the OBJECT always on liquid

↓ UPSIDE DOWN ⇒ NOT solid goes down in the VAT, instead the object attached on top, elevator move up the polymer

Introduction to Stereolithography

SL create
strong boundaries

extreme complex object
tough resin objects



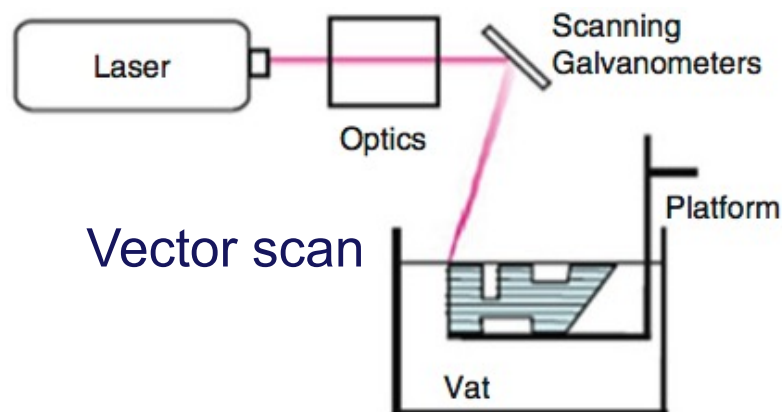
↓
small
VAT, less
liquid,
scan
from below
with
transparent
VAT

<https://youtu.be/yW4EbCWaJHE?list=RDyW4EbCWaJHE>

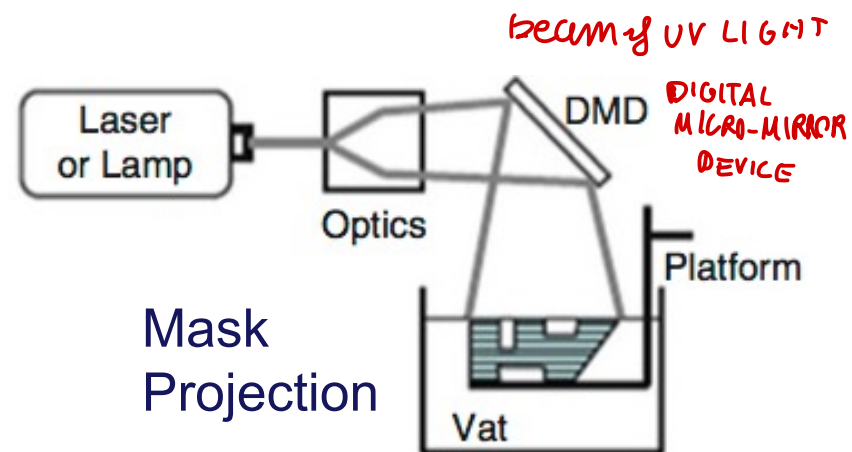


VAT PHOTOPOLYMERIZATION

Schematic diagrams of three approaches to photopolymerization processes. *further evolution*

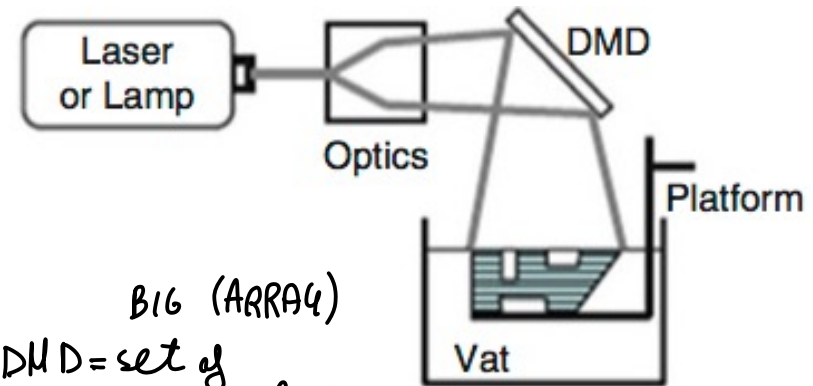
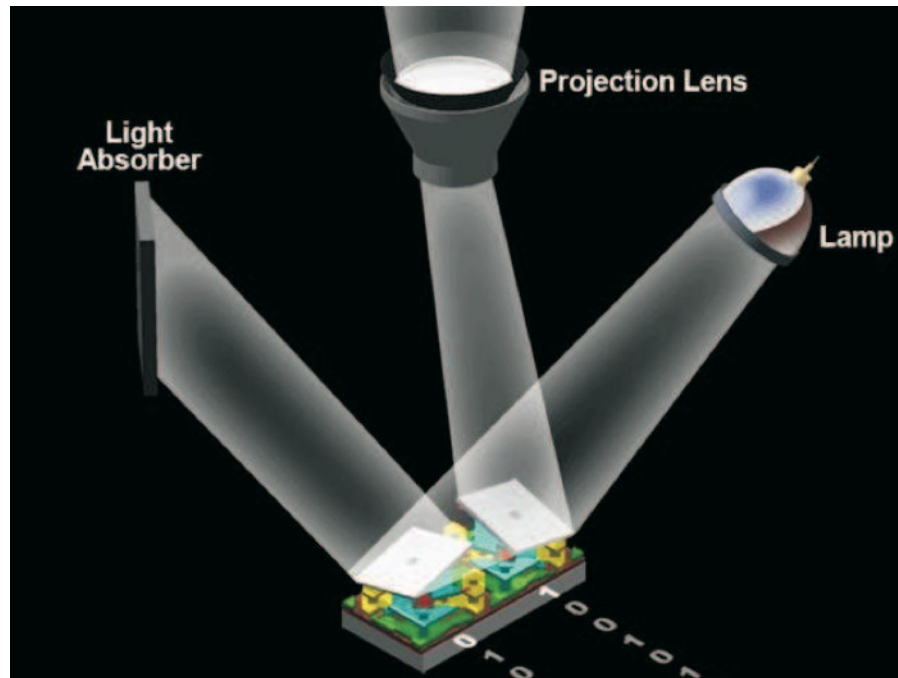


DMD reflect light,
DIGITAL cause here you
have small mirrors
directing the light,
CONTROL DMD



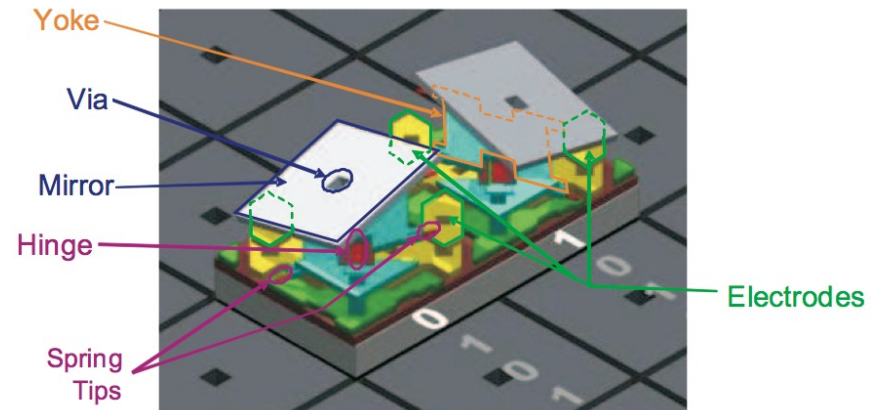
*Why to scan and not send a
surface image?
UV white image of layer to
solidify? MASS- PROJECTION*

Mask Projection VP Machines: Mask Projection Stereolithography (MPSL)



BIG (ARRAY)

DHD = set of
MIRRORS controlled in position
JUST POSITION MIRRORS



each MICRO-MIRROR with distance give the resolution

RESOLUTION you can give to surface

you can send an image to surface

{ IF BIG Sugar is better to scam }

← applied for int so
big surface

CONTROLLING the focus of the scan is different between

SCAN BEAM and MICRO-MIRROR

constant
resolution

depending on
distance → high / low
accuracy

Full Image ↓

polymerization easy

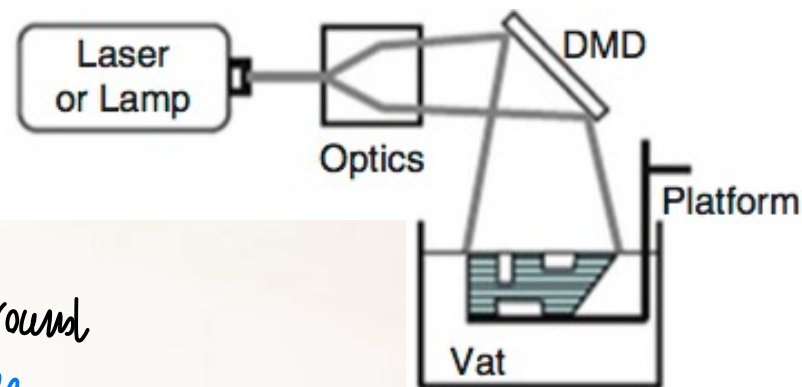
DMD can be used upside-down

+ illuminate from below → bigger
print



VAT PHOTOPOLYMERIZATION

Mask Projection VP Machines: Mask Projection Stereolithography (MPSL)



define the steps to correct the position
and then form the plastic to shape the material around
uv light polymerize plastic liquid resin in required area ...
exposed surface, it goes down
hybrid/ceramic material for prototype .. etc

Automotive
DENTAL → transparent "upper arch" for dental position
personalized part
SCAN MOUTH, create 3D shape and create the geometry, steps for teeth correction
print physical changes of teeth
thermoform the plastic sheet to shape material around mouth

AM, SL to create a personalized product!

<https://youtu.be/l104iZ2w4zg>



AM PROCESS CATEGORIES

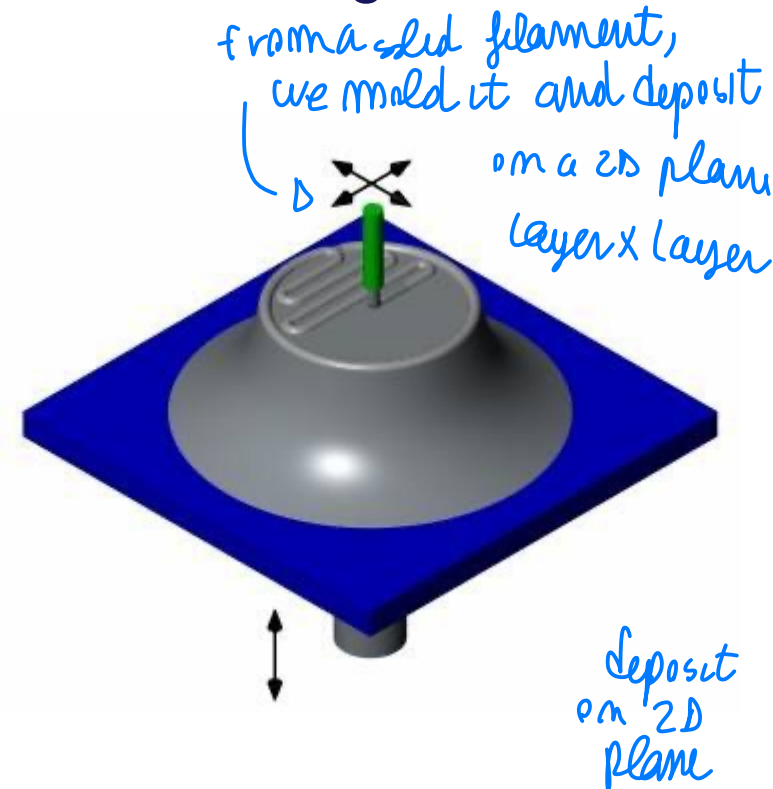
[cheapest way to create parts]

heat the solid until melt and let it pass through a die shaped

Material extrusion: an additive manufacturing process in which material is selectively dispensed through a nozzle or orifice.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.



IT IS AN EXTRUSION process → from a solid thermoplastic → heat → polymer melt
 we use a NOZZLE with CIRCULAR CROSS-SECT. → let material flow and deposit it controlling position of NOZZLE

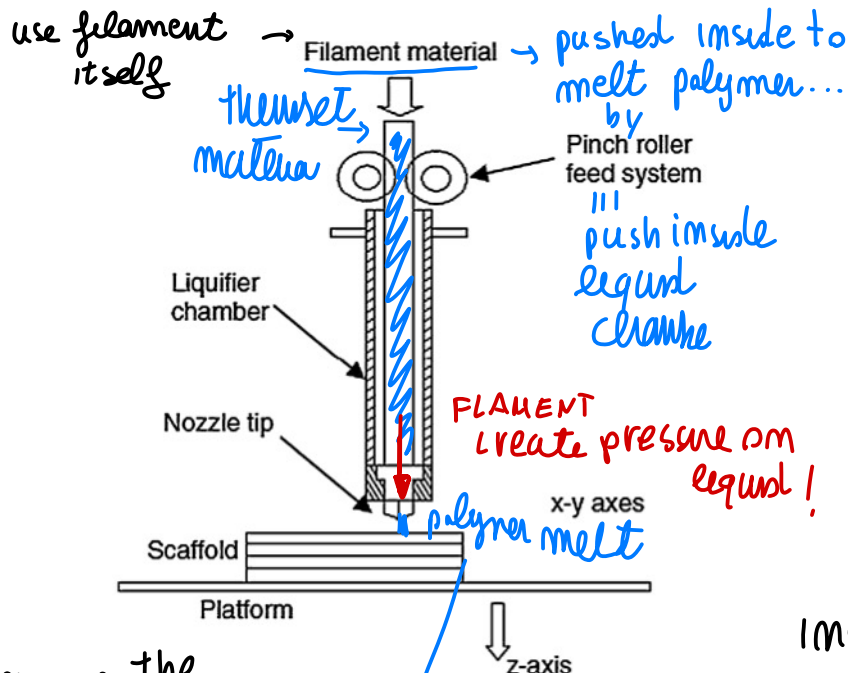
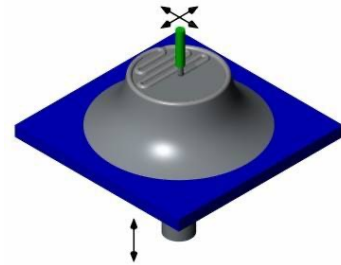


MATERIAL EXTRUSION

28

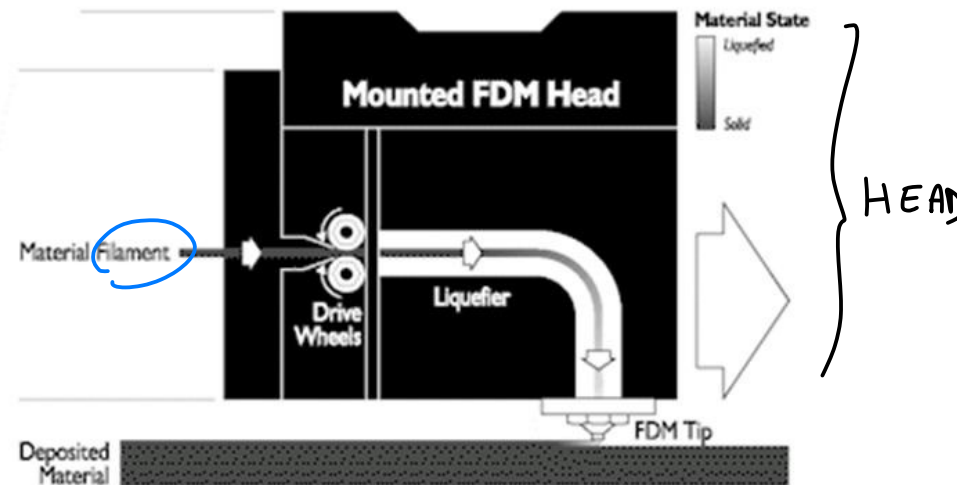
A continuous filament can be pushed into the reservoir chamber, thus providing a mechanism for **generating an input pressure for the nozzle.**

the HEAD is the important part!



you use the filament to push through nozzle

Solid enough to deposit

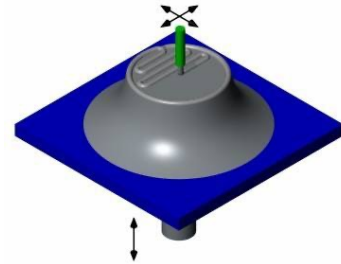


Inside the HEAD you heat material until liquid viscous through the nozzle



The machine must be capable of:

- ^{2D} scanning in a horizontal plane (controlling the workhead in x-y plane during each layer);
- starting and stopping the flow of material while scanning; *CONTROL deposition → selectively deposit*
- moving upwards by a distance equal to one layer in the z-direction (or the part downwards);



Part is fabricated layer-by-layer from the base up.



MATERIAL EXTRUSION

YOU CAN HAVE SMALL MACHINE
very simple JUST controlling AXIS
and elevator + filament plastic

30

→ LIGHT COST!!

respect VAT Photopolymeriz.

An additive manufacturing process in which **material** is **selectively dispensed** through a nozzle or orifice.

you can have more than one HEAD to manage more materials...

(to avoid damaging the material...)

MATERIAL EXTRUSION

LOW COST
PROCESS

one material for
support, one for the
part. so easy

support to remove
OR to create multi
material part

• FDM Fused Deposited Material

(Layer additive process → excellent strenght
+ VIBRATION) Build layer by layer

extrusion nozzle position the material continuously

to move on a planar 2D → hands handy for surface finish

(AEROSPACE / AUTOMOTIVE / MEDICAL ETC)

→ 2 head nozzle to add
different material

• FFF Fused Filament F

UV necessary
COST LASER

(SL) ↑
VAT polym.
very accurate

--- ↓ ---
• here it is
less accurate,
less thickness
if layer
depending on
nozzle
material.

faster and
less COST
Just polymer heat

<https://youtu.be/WHO6G67GJbM>



Binder jetting: an additive manufacturing process in which a liquid bonding agent is selectively deposited to join powder materials.

Materials	Example materials	Process categories						
		✓ Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

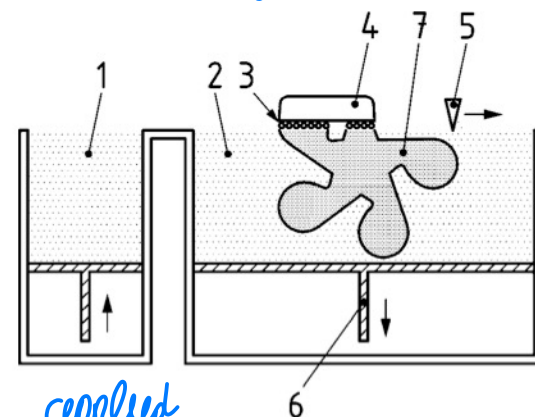
Note: Combinations of the above material classes, e.g. a composite, are possible.

↓ Classical Jetting technologies comes from a 2D printer Jetting

↓ just printing a plane of additive (glue)

↓ {to bind the polymer}

selectively deposit the glue to link together



applied on many materials

Key

- 1 Powder feeding system
- 2 Powder material distributed in a powder bed
- 3 Liquid bonding agent
- 4 dispensing apparatus including connection to bonding agent feed system
- 5 Powder spreading device
- 6 Build platform and elevator
- 7 Product

link together different parts of material ...



BINDER JETTING

↪ this technologie is quite old...

(one of the FIRST Additive manufacturing process)

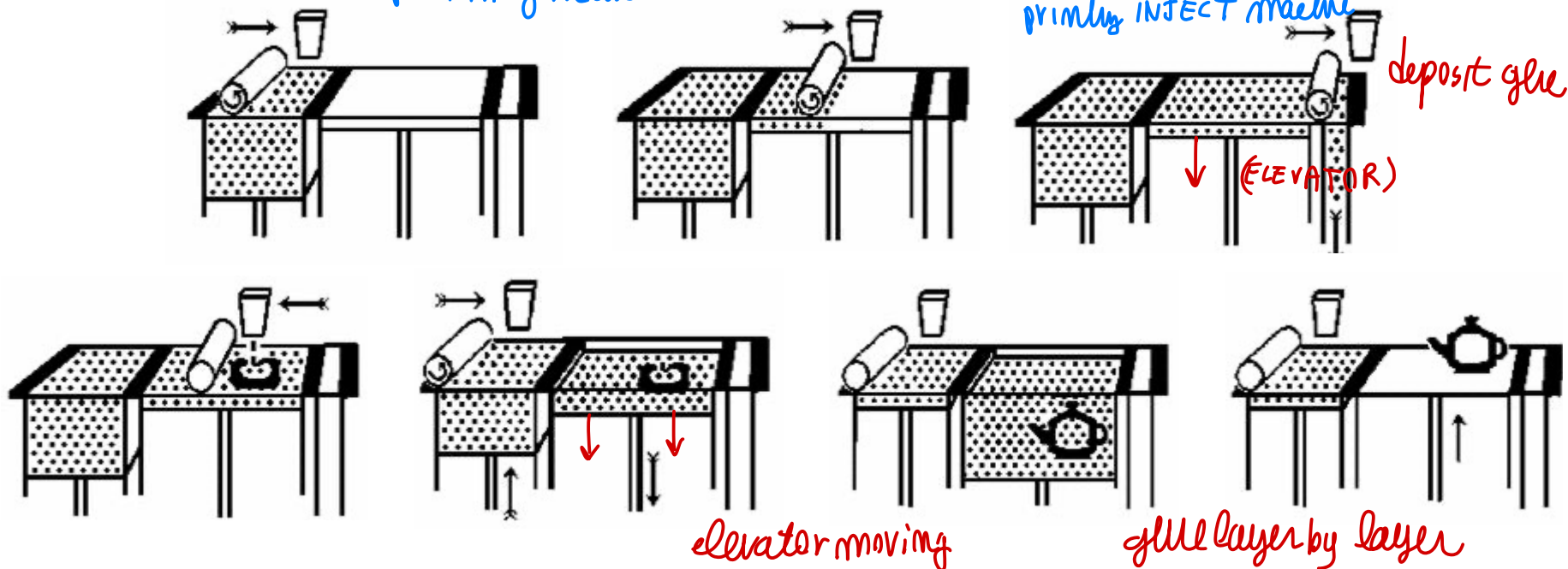
32

Binder jetting methods were developed in the early 1990s, primarily at MIT. They developed what they called the 3D Printing (3DP) process in which a binder is printed onto a powder bed to form part cross sections.

instead of printing the INK,
let drop glue or material outside
printing head



machine similar to
printing INJECT machine



Simple logic $\rightarrow$ material distributed... Linked together

move down the elevator of thickness $\Rightarrow$ to continuous iterate
until obtained the
solid object

LOGIC

distribute a layer of printing material (from elevator)

use print head to deposit the GLUE on link selectively

the powder on each layer $\rightarrow$ glue each layer and

move down elevator

related to layer thickness

$\Downarrow$

distribute powder

(continuous iteration $\rightarrow$ obtained object by removing
non used powder $\rightarrow$ (reusable for
further process))



BINDER JETTING

this process can bring to a final OBJECT
functional object

33

When Z Corporation first started in the mid-1990s, their first material was **starch** based and used a **water-based binder** similar to a standard house-hold glue. *(at beginning it was used JUST to make prototype, @ beginning we used NON FUNCTIONAL material)*

At present, the commercially available powder

- **plaster based** (calcium sulfate hemihydrate) with water-based binder; *usually now we use polymers (as glue and powder) → polymer-based PART*
- **PMMA** (poly-methyl methacrylate) powder and uses a liquid binder that reacts at room temperature;
- **wax-based binder** for use with PMMA powder to form pattern for investment casting.

you may use METAL + adhesive → gluing together metal powder → for relevant elements

With either the starch or polymer powders, parts are typically considered visual prototypes or light-duty functional prototypes.

Infiltrants are used to strengthen the parts after they are removed from the powder bed. In some cases, particularly with elastomeric infiltrants, **parts can be used for functional purposes.**



Composite material → with different kinds of material

GLUE ≈ polymer to be CURED + INFILTRANCE (use additional liquid material to fill the gaps and make material stronger)
(to make STRONGER) Use

get a
composite
Material

↓
⇐ you substitute the GLUE with an
INFILTRANCE (LOW MELTING POINT metal)

ISSUE: you cannot have closed cavity or powder remain inside

you can use the polymer base to shape the METAL component

with low melting metals → composite
metallic material from a thermoplastic base

↓ obtain a composite
metallic part



AM PROCESS CATEGORIES

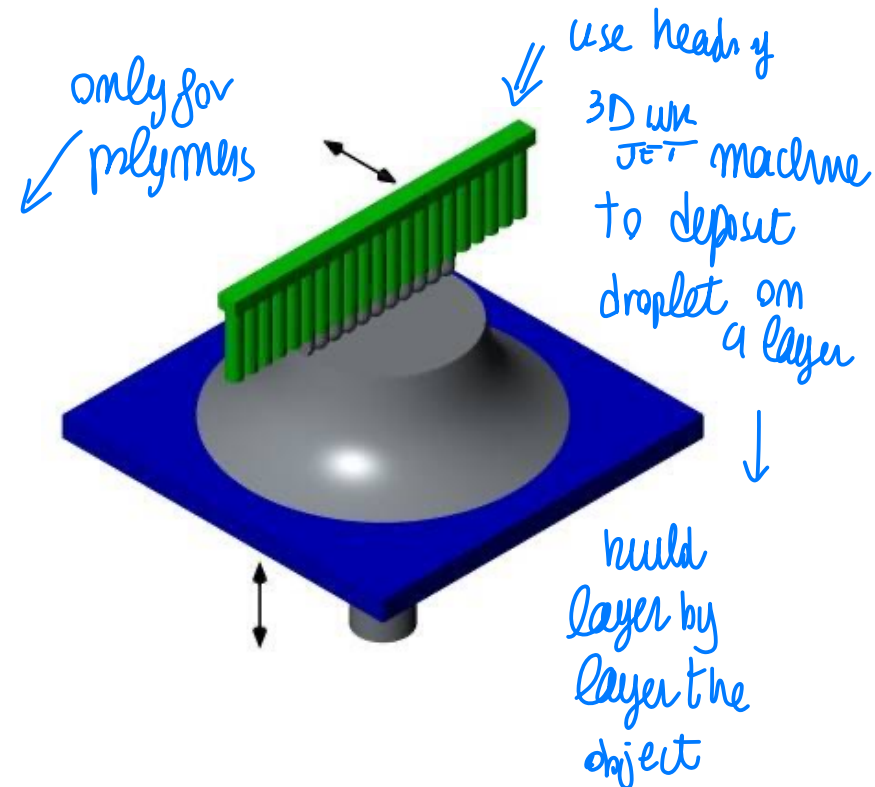
it is an evolution of

similar to BINDER JETTING, conceptually different → BinderJET

Material jetting: an additive manufacturing process in which droplets of build material are selectively deposited.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.



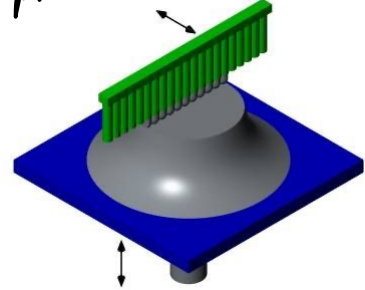


MATERIAL JETTING

from thermoplastic - you apply heat
polymer melt through NOZZLE (ARRAY of nozzles)

36

smaller nozzle → better detail but more pressure needed



Printing technology has been extensively investigated, with the majority of that investigation historically based upon applications to the 2D printing industry (inkjet printing has been in existence since the 1960s).

Printing as a 3D building method was first demonstrated in the 1980s with patents related to the development of *Ballistic Particle Manufacturing*. The first commercially successful technology was introduced in 1994, which printed a basic wax material that was heated to liquid state.

It is notable that the first generation of 3D printing machines relied on heated **waxy thermoplastics** as their build material; they are therefore most appropriate for concept modeling and investment casting patterns.

More recently, the focus of development has been on the deposition of **acrylate photopolymer**, wherein droplets of liquid monomer are formed and then exposed to ultraviolet light to promote photopolymerization.

evolution by using this polymer that can be cured by UV light

big set of elements with small nozzle $\equiv$ high pressure

WAXY THERMOPLASTICS $\rightarrow$ 3D printing as PATTERN for INJECTION CASTING!

$\rightarrow$ to have a functional parts, with high performance

$\downarrow$

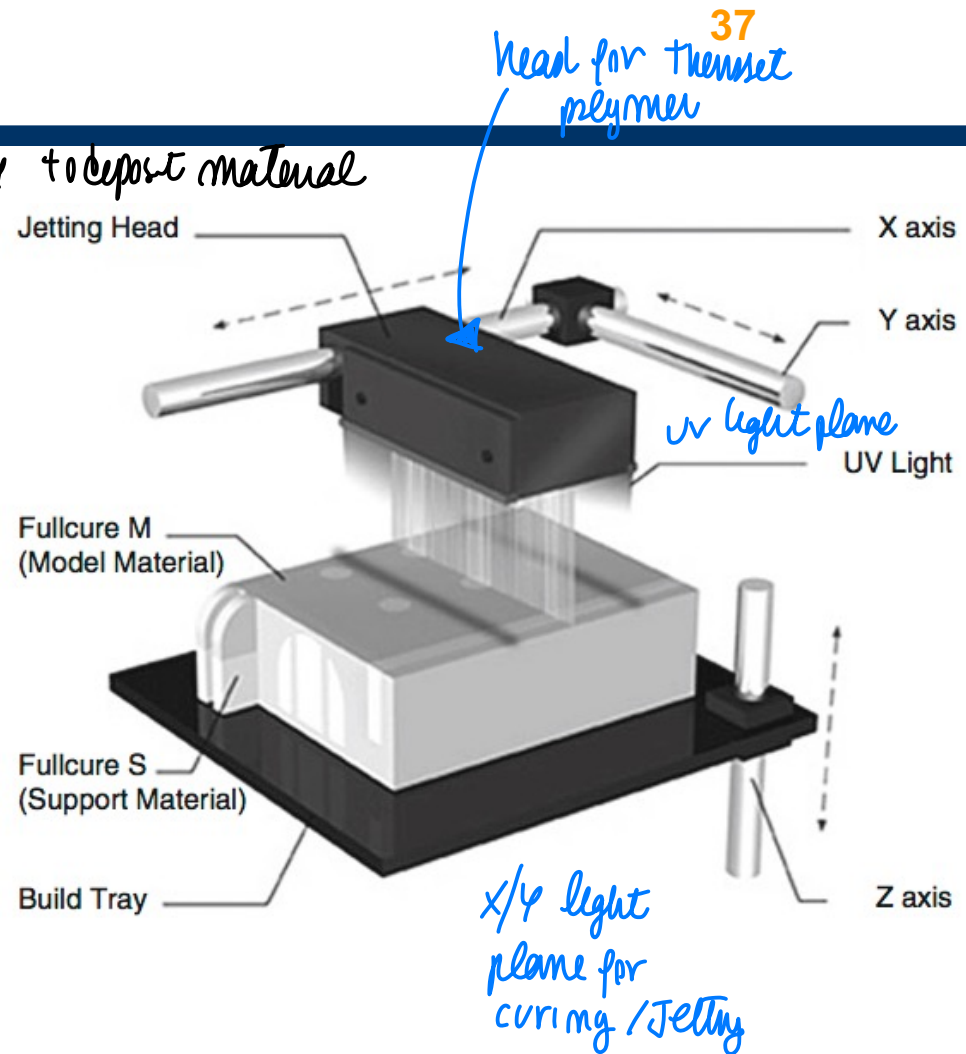
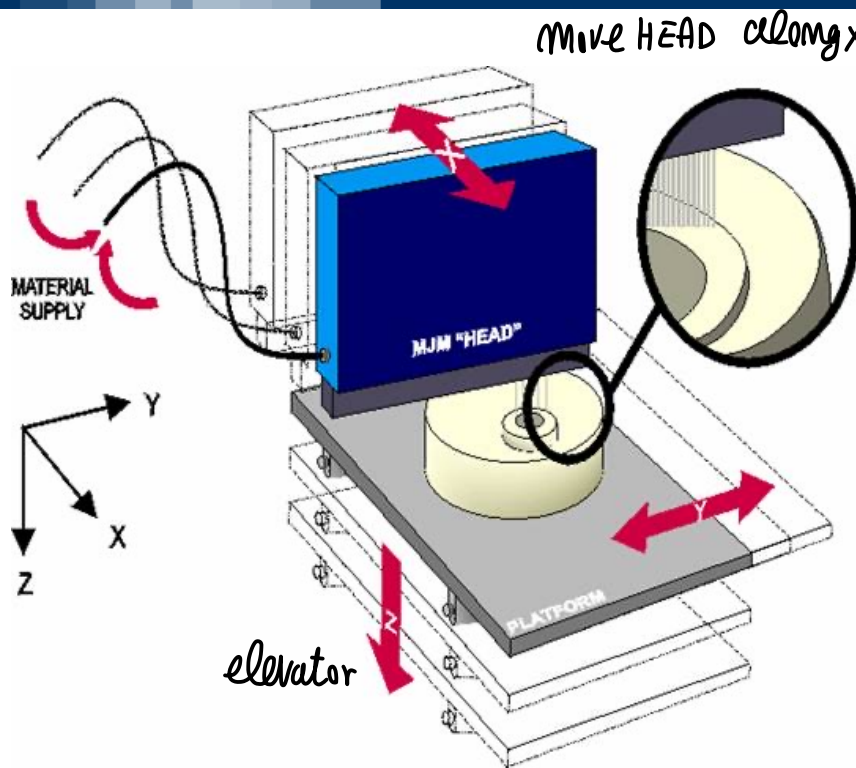
this process use ACRYLATE PHOTOPOLYMER

$\downarrow$

scan layer with UV light
to solidify the layer



MATERIAL JETTING



Droplet Formation:

- Continuous Stream
- Drop on Demand



{PolyJET} Jet the resin which react with UV light
Build CROSS-SECTION layer by layer
smooth parts With good looking detail
(HEAD is the Core of technique)

"OVERMOLD" MATERIAL → multimaterial polymeric part Just
by managing multiple material thanks to multiple
Nozzle on the head
so easy creation of supports easy to remove from final object
⇒ different kind of material on same part
(like in OVERMOLD INJECTION MOLDING)

Detail related to nozzle dimension
cause you have more nozzle
leakage droplet problem! ⇒ Decide from which nozzle set ejecting the polymer melt → CONTROL nozzle ejection
↖ [V Array nozzle]
{ by streaming droplet & nozzle... } selectively choose which droplet goes to part or droplet on demand → different tech



Primary advantages include:

- **low cost** (no use of laser), simple technology, low cost
- **high speed** (from hundreds to thousands of nozzles), → FAST
- scalability (adding print heads),
- ease of building parts in multiple materials,
- capability of printing colors.

usually we don't need supports --- the powder itself
create support with the other object parts

compared with binder Jetty → here is better and easy to create supports

on binder Jetty, powder

itself create support of

other object layer → vat full of
powder with object

→ usually NOT necessary support
ON BINDER JETTING

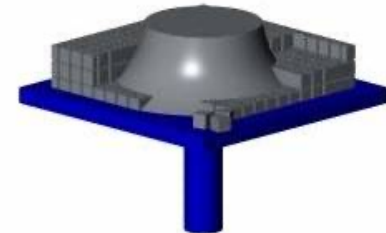
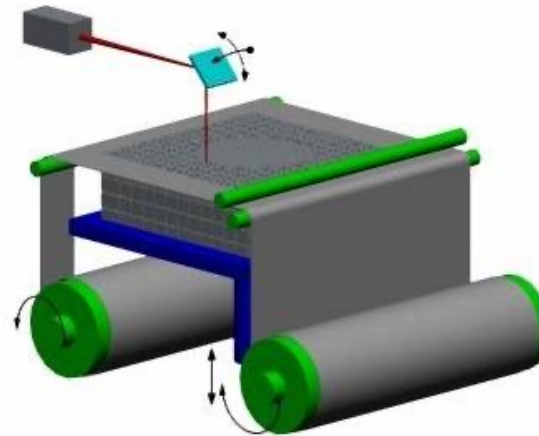


Sheet lamination: an additive manufacturing process in which sheets of material are bonded to form an object.

One of first technologie → LAMINATED
OBJECT MANUFACTURING 1991

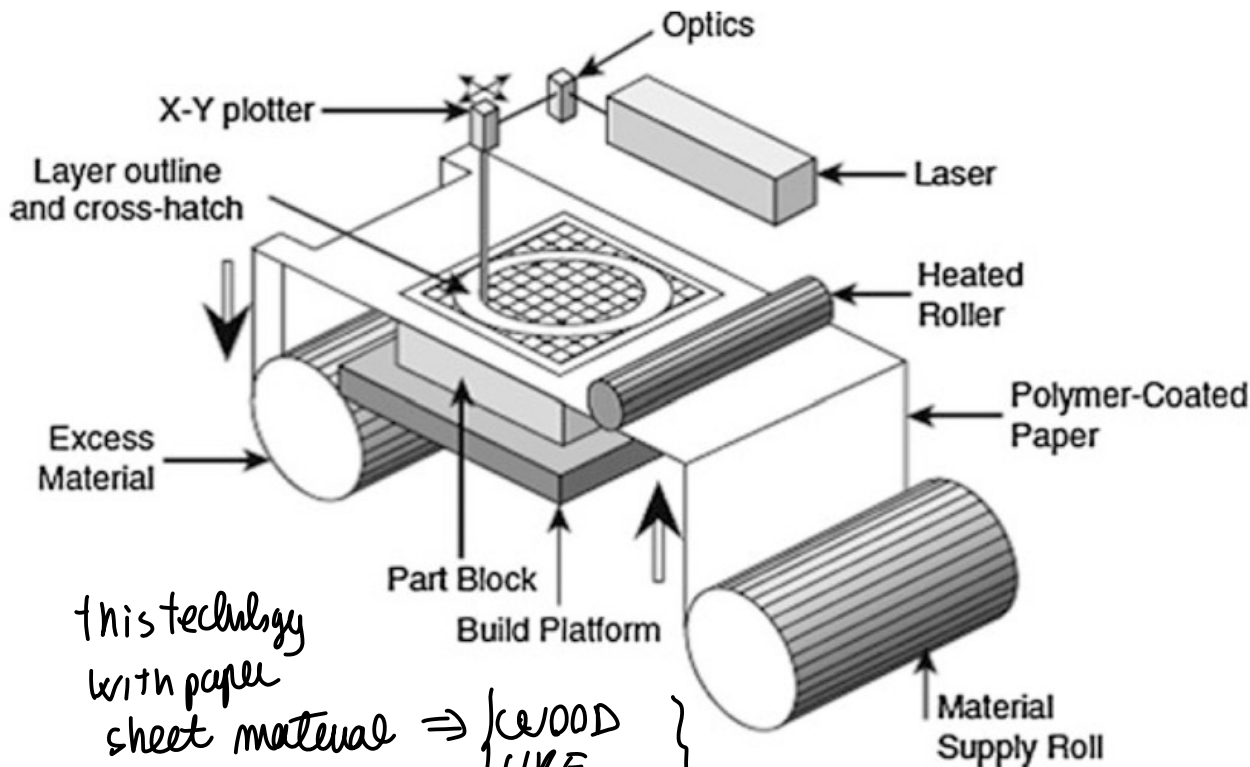
Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.





One of the first commercialized (1991) AM techniques was Laminated Object Manufacturing (LOM).



this technology
with paper
sheet material $\Rightarrow$ {WOOD
LIKE
MATERIAL}

usually use
for PATTERN creation for SAND CASTING
 $\rightarrow$ for the future metallic part by molding

LOM involved layer-by-layer lamination of paper material sheets, cut using a CO₂ laser, each sheet representing one cross-sectional layer of the CAD model of the part.

In LOM, the portion of the paper sheet which is not contained within the final part is sliced into cubes of material using a crosshatch cutting operation.

starting from a continuous sheet of material...

from full layer (polymeric adhesive → with an HEATED ROLL

CUTTING the material

for ex with LASER technology

compressing and bonding the layers!)

→ well focus!

LOGIC..

until now just selectively deposit / cure material,
single element of material layer by layer

↑
NOT CONTINUOUS material

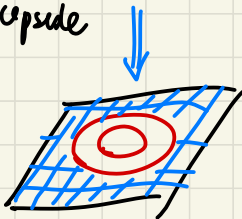
here we start from a sheet of material
(@ bagging paper with adhesive)

then cut
boundaries & layer

CUT for ex with LASER tech

move cutting laser on x-y

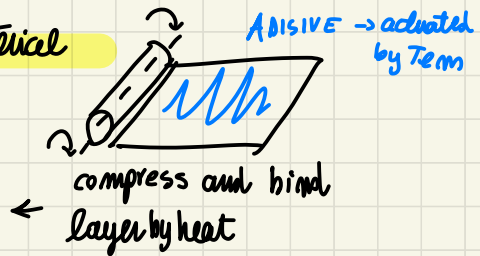
+ more CUTS on layer
cause if I hand new layer
upside



cut (—)

IF single layer

glue → unique Solid NOT adapt for
a complex shape...
not able to extract real part



With The correct focus
calibration to NOT defect
the layers below

↑
ACCURATE on
right focusing



In some processes the laminate is bonded first to the substrate and is then formed into the cross-sectional shape (“**bond-then-form**” processes).

cut layer and
then placing
a new layer,
with adding cuts
to **early** remove
later ↑
additional cut
necessary to remove
material NOT PART
of your object



<https://youtu.be/aAbBvrfS6TY>



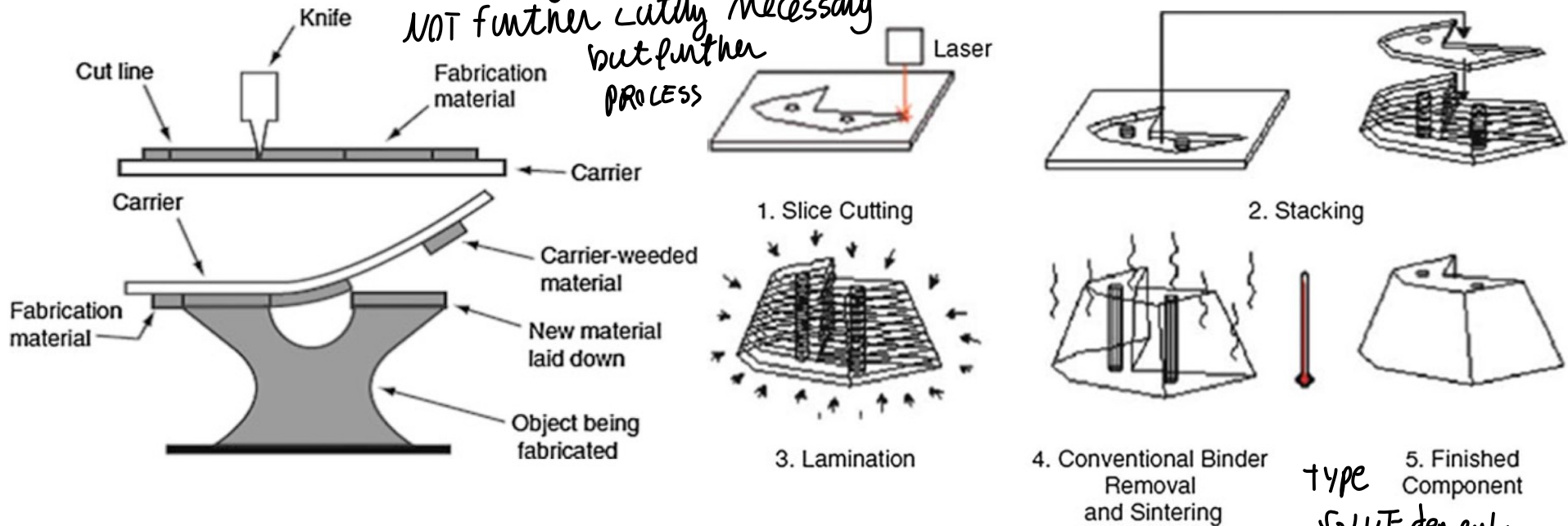
SHEET LAMINATION

2nd way to do lamination

43

In **form-then-bond** processes, sheet material is cut to shape first and then bonded to the substrate. This approach is popular for construction of parts in metallic or ceramic materials that are thermally bonded.

Welding technologies necessary to link the parts
NOT further cutting necessary
but further
PROCESS



type
GLUE depends
on material

Offset fabbing system

CUT the shape before and then Bond the
different layers
sample sheet → CUT → shaping that layer
↳ assemble the layer and glue it together

CAM-LEM Process

Computer-Aided Manufacturing of
Laminated Engineering Materials

according to material



SHEET LAMINATION

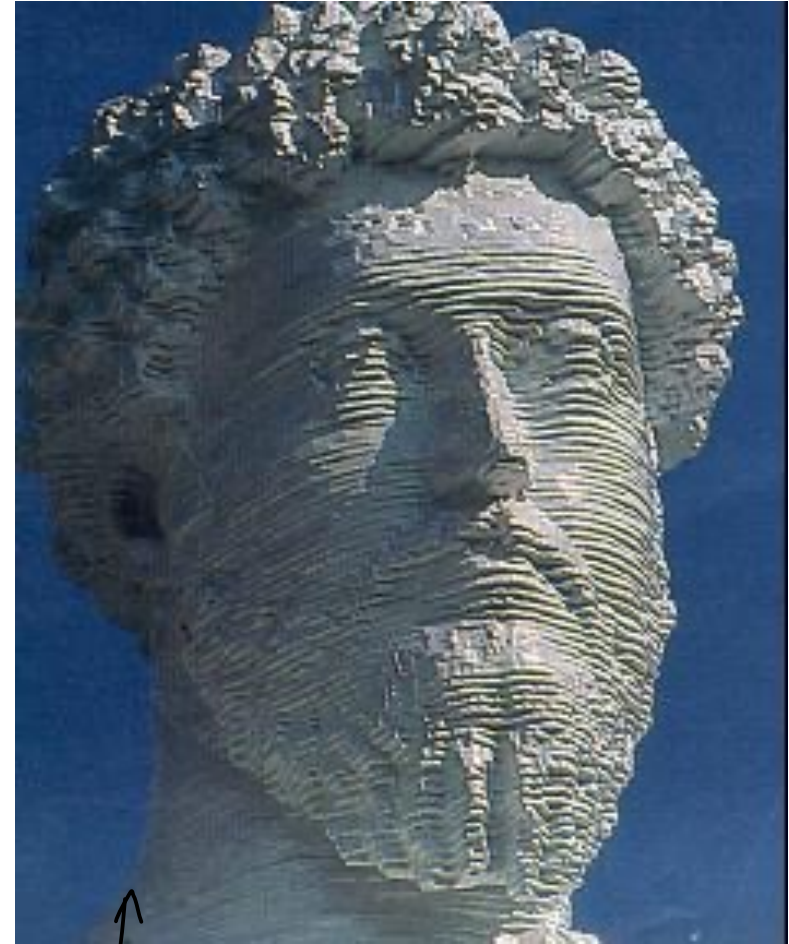
FOR BIG OBJECTS → we have few technologies available BIG OBJ → challenge 44

GOOD FOR BIG OBJ
(also by SAND CASTING)



Copy of the STATUE made of polystyrene

190



The Equestrian Statue of Marcus Aurelius
in the Campidoglio, Rome, Italy
here from because the layer is homogeneous

different layer assembled → then you will finish surface

for a BIG OBJECT... multiple processes
challenge of building → difficult!

issue → Dimension → can be done by sheet lamination

layer by layer → usually you have nonhomogeneous material
issue along growing direction

↓
true & process

↑ it can be solved
by post processing
(well known problem)

With sheet
lamination → layer homogeneous

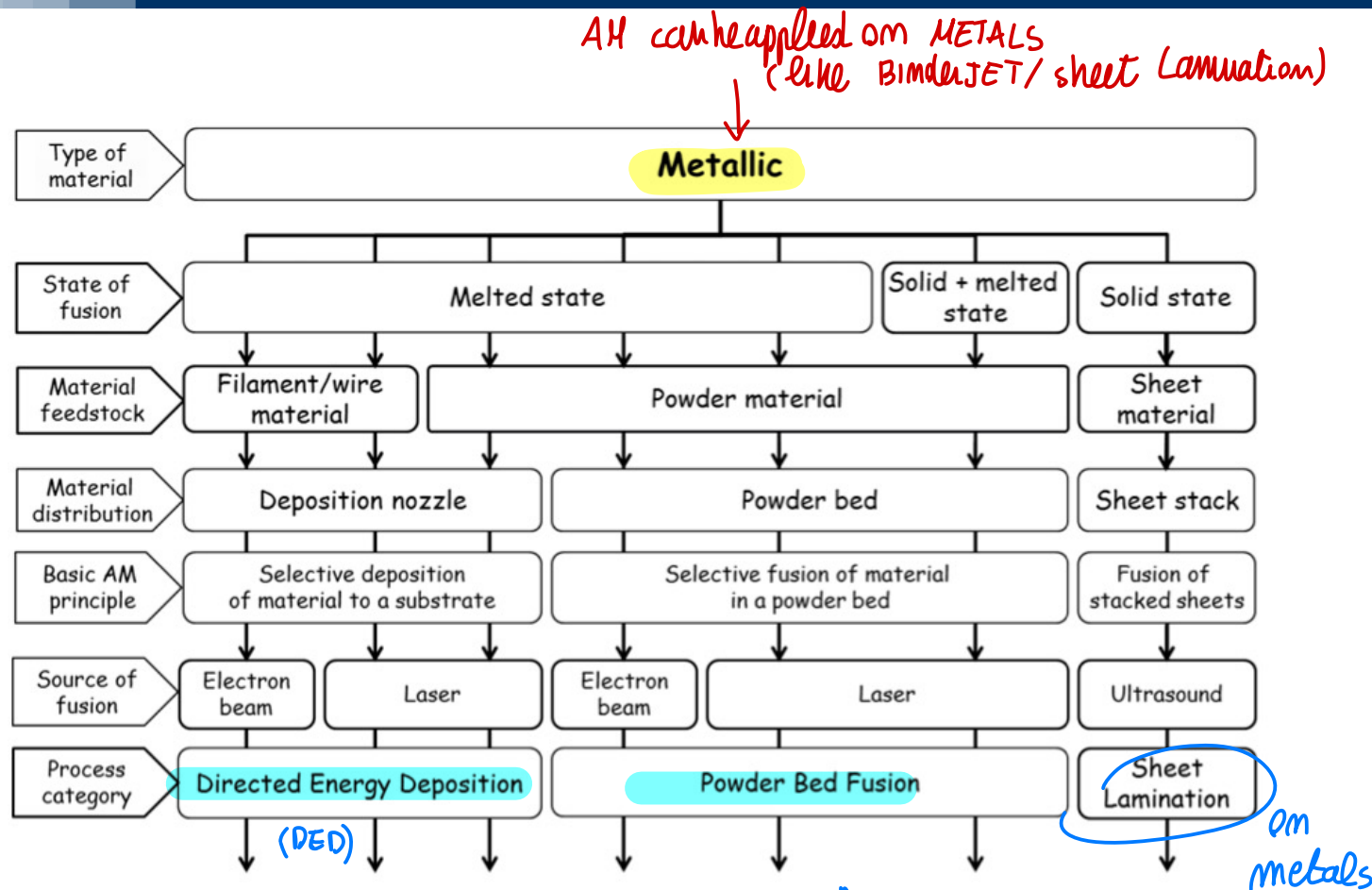
where another process also on same layer is NOT homogeneous

the removed material, depending on
what kind of material is used sometimes → recycle problem
it can be reused others can't be and is wasted

(Difficult but possible)



AM PROCESS CATEGORIES



on the schema we say different category

↑ Metal
processes that we
will describe



AM PROCESS CATEGORIES

Powder bed fusion: an additive manufacturing process in which thermal energy selectively fuses regions of a powder bed.

from a powder layer → INSTEAD of glow together → link the particles of powder to make a 3D obj

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.

METALLIC MATERIAL



try to link together particles of the powder to create
3D objects



POWDER BED FUSION: PROCESSES

↓ so different ways of linking / lot technologies

Then, according to the **energy source** and the **fusion mechanism**,

there are: CREATE the layer selecting the material

from powder (NOT JUST Metal but METAL covered but Adhesive binder)

similar material binder for investment shell casting

melt material to link element

(here meet more)

- **Selective Laser Sintering (SLS):** Laser beam and liquid phase sintering

metal covered with a binder (using laser) to melt binder to link the powder

- **Selective Laser Melting (SLM):** Laser beam and full melting

HEATING powder selectively to create layer and link it, just that metal
LASER bring high ENERGY (higher than VATOPHOTOPOL..) high melting point metal

- **Electron Beam Melting (EBM):** Electron beam and full melting

vacuum chamber → with ELECTRON BEAM that scan the powder bringing energy for melting

to have a full uniform unique material

full METAL

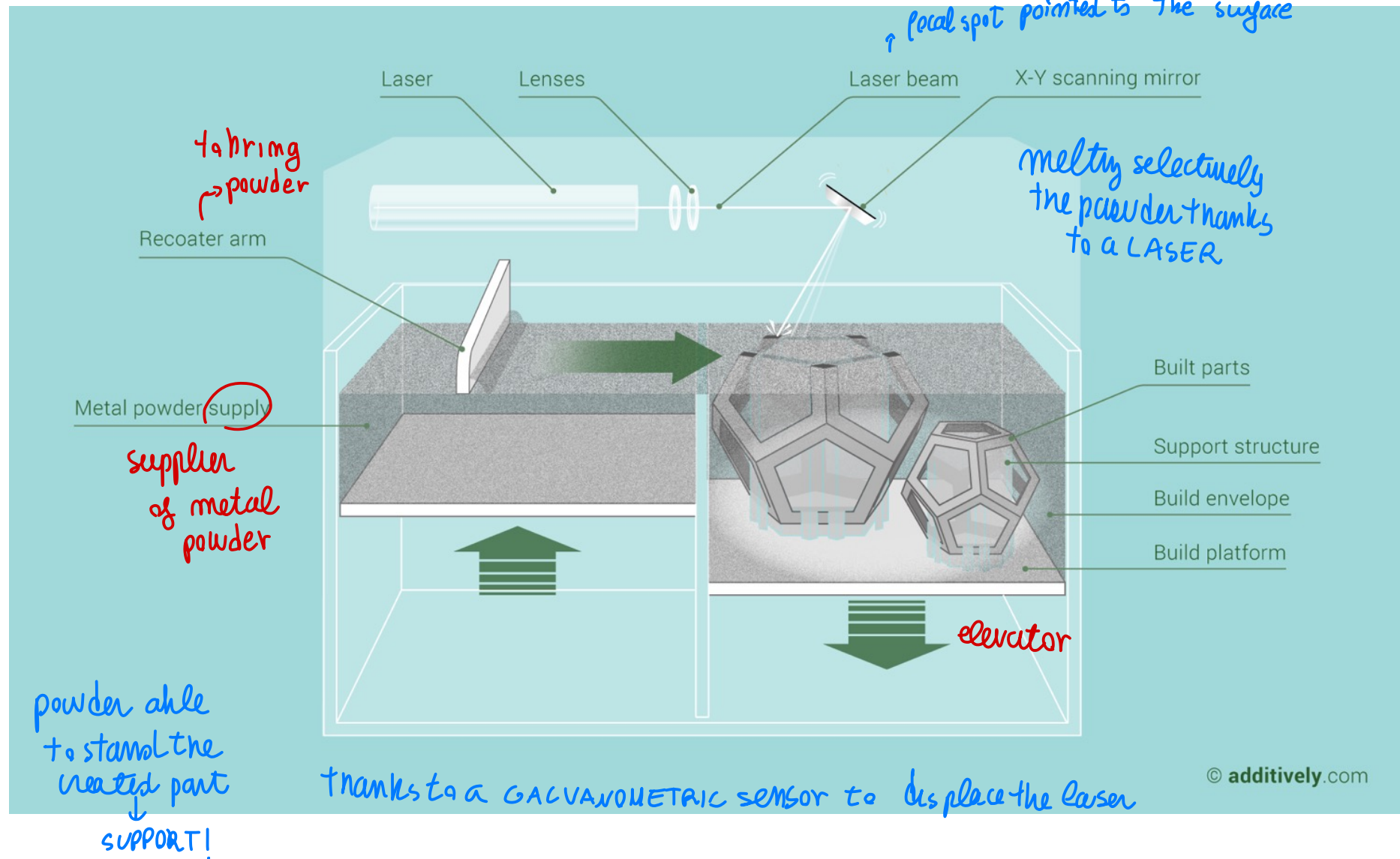
MATERIAL @ HIGH melting point



(Schema similar to BINDER JETTING)

Melt selectively the powder...

SELECTIVE LASER MELTING



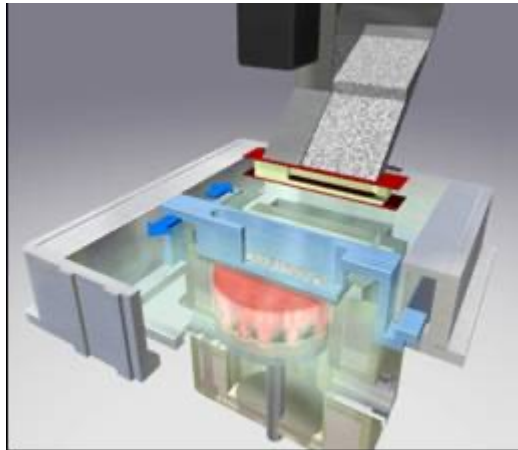


SELECTIVE LASER MELTING

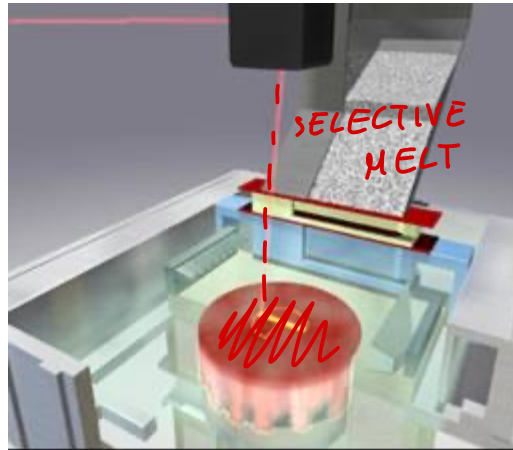
selectively melt the material

red color → cause metal powder preheat so

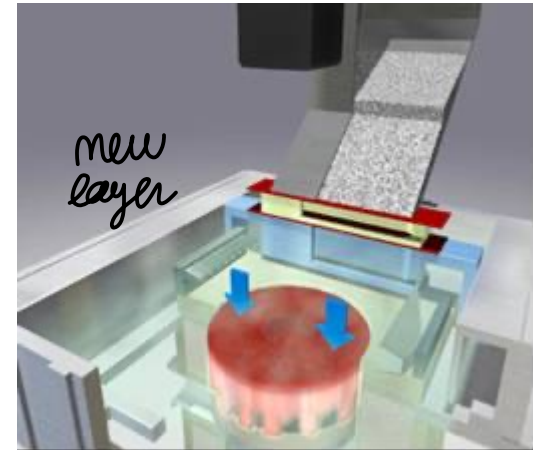
less energy to transform



1) New powder layer

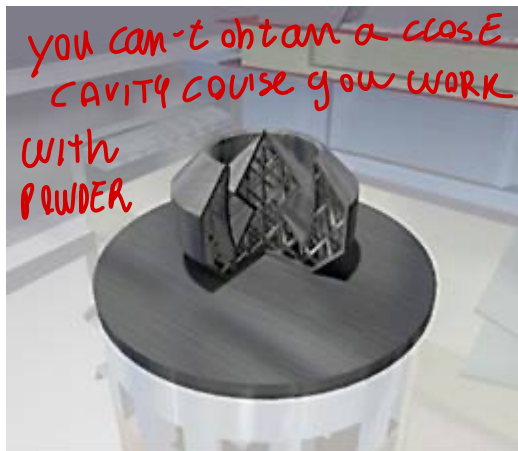


2) Powder melting

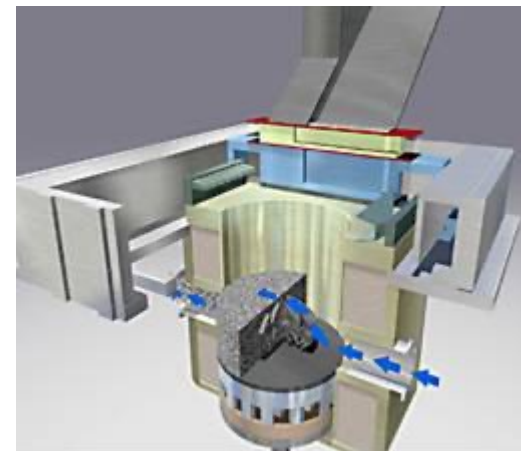


3) Platform lowering

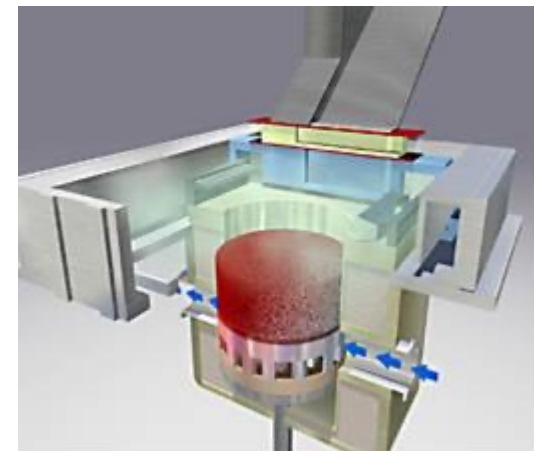
remove powder + tanks to open holes



6) Part final



5) Extra powder removal



4) Cooling down

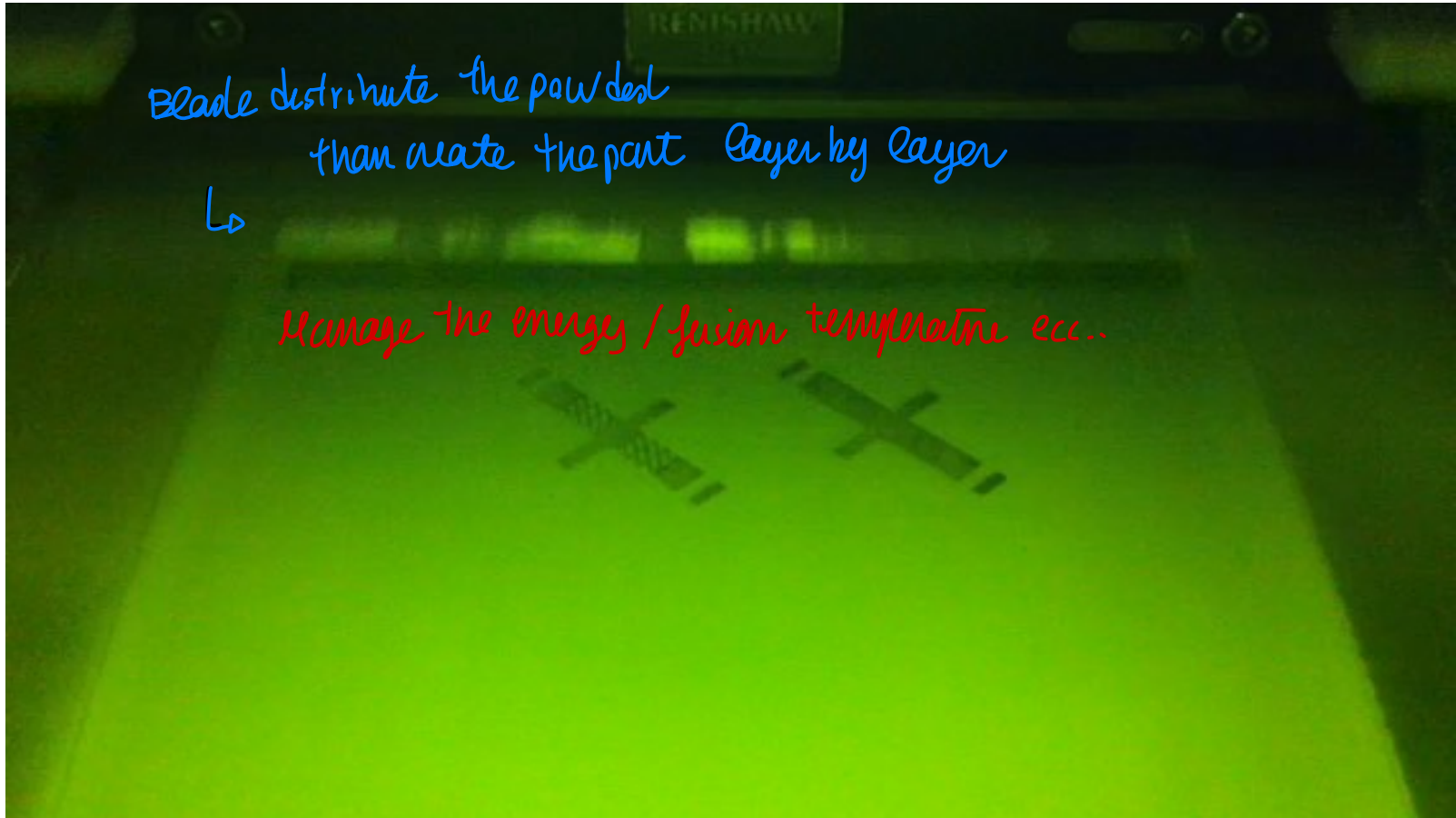


SELECTIVE LASER MELTING

Blade distribute the powder
than create the part layer by layer

↳

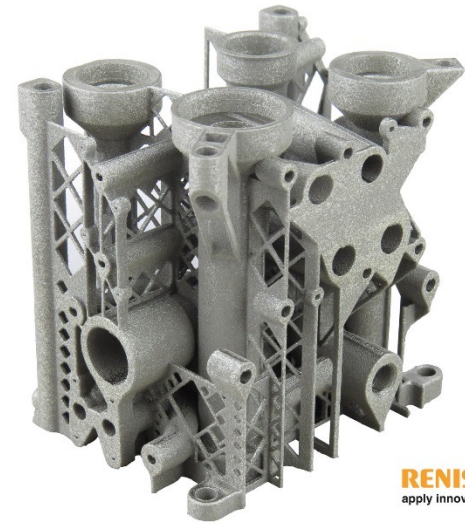
Manage the energy / fusion temperature ecc..





SELECTIVE LASER MELTING

you can create complex metal geometry



RENISHAW
apply innovation™

RENISHAW
apply innovation™

Development manifold produced on the Renishaw AM250 demonstrates how complex structures can be additively manufactured.

REF: REN013

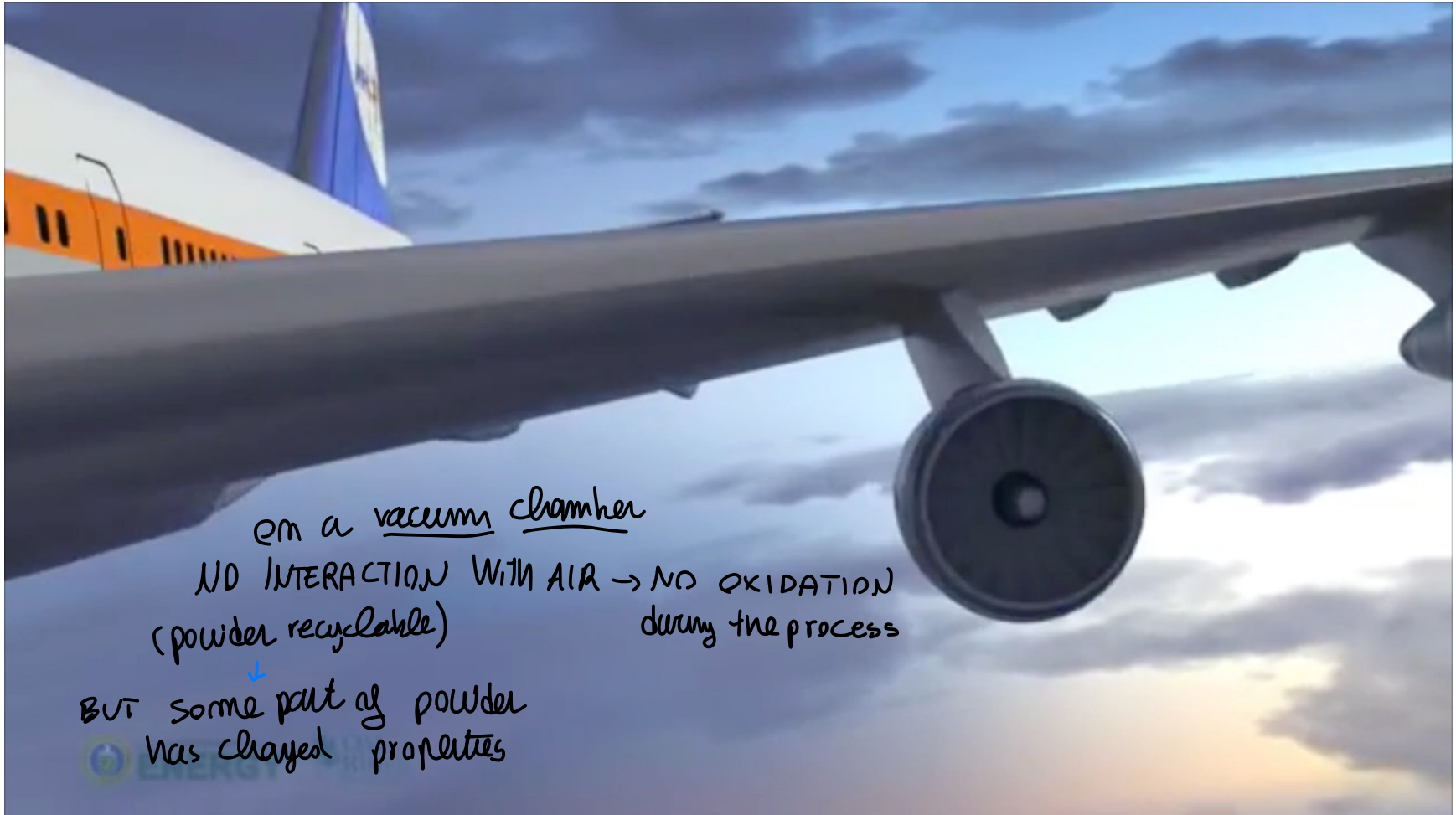


Images courtesy of Renishaw



ELECTRON BEAM MELTING

applied a lot in AEROSPACE
(ex: for a turbine blade)



Sorce: Arcam



AM PROCESS CATEGORIES

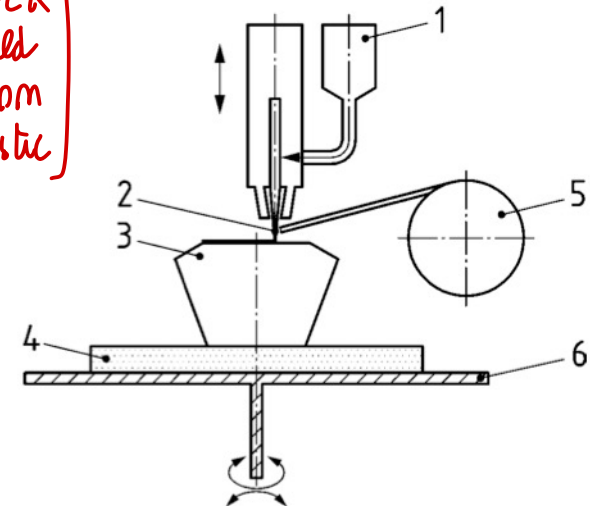
Directed energy deposition (DED): an additive manufacturing process in which focused thermal energy is used to fuse materials by melting as they are being deposited.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxy and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.

(While POWDER BED could be applied on thermoplastic)

ONLY ON METALS!



Key

- 1 Powder hopper
- 2 Directed energy beam, for example: laser, electron beam or plasma arc
- 3 Product
- 4 Substrate
- 5 Wire (filament) coil
- 6 Build table

using a different technology: POWDER or SOLID STATE metal different source to melt and add material needed



DIRECTED ENERGY DEPOSITION

*different ways to
do this...*

According to the ^{*different*} **energy source** utilized one can distinguish:

- **EBMD – Electron Beam Metal Deposition**
- **LMD – Laser Metal Deposition**
- **PMD – Plasma Metal Deposition** ← *PLASMA = classical melting
technologie → create intense melt*

According to the ^{*different*} **raw material type** one can distinguish:

- **Powder Feeding**
- **Wire Feeding** → *SOLID WIRE MATERIAL*



According to the **feeding modality** one can distinguish:

- **Coaxial Feeding**
 - **Off-Axis Feeding**
- respect the energy source!*

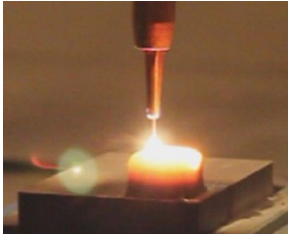
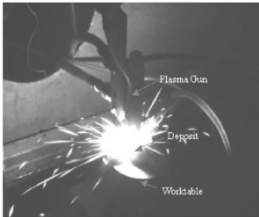
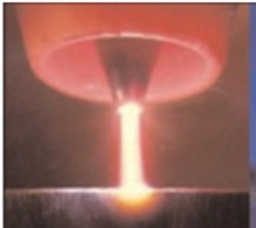


DIRECTED ENERGY DEPOSITION

deal with..

↓ POSSIBLE COMBINATION

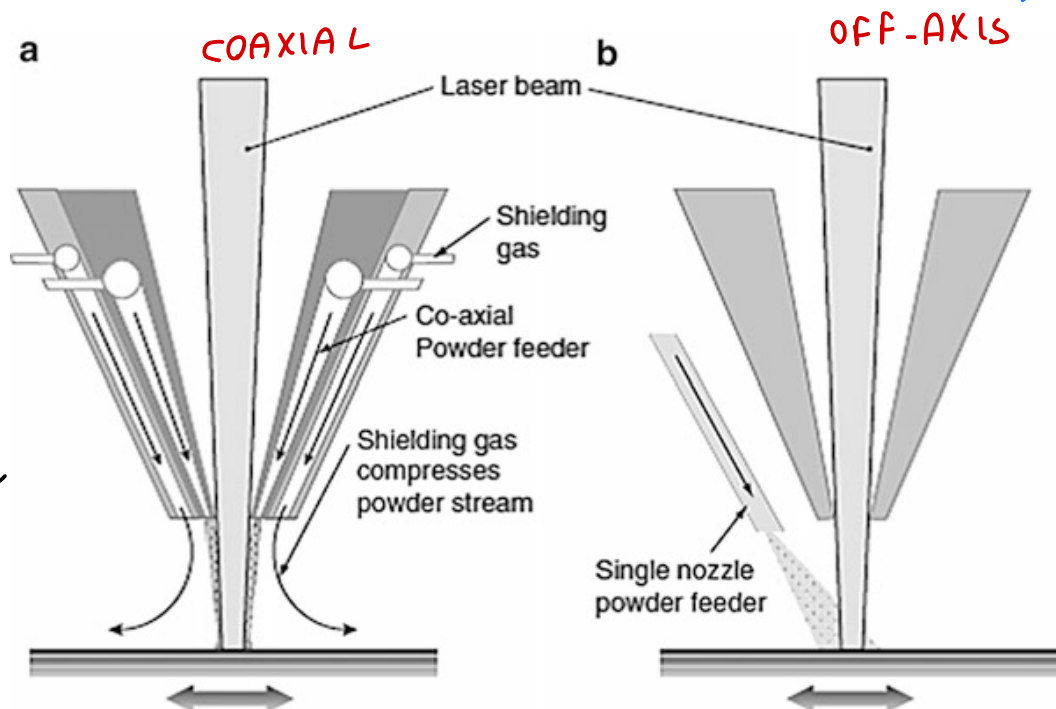
only
WIRE ←

	Powder	Wire
Laser	Laser metal deposition 	Laser wire deposition 
Electron beam	n/a	Electron beam wire AM 
Plasma/arc	Plasma deposition manuf. 	Plasma, arc wire feed AM 



LASER METAL DEPOSITION: POWDER FEEDING

POWDER / WIRE
feeding of
the material
↓
we add just
the needed
material
add selectively the
needed material and
melt positioned
material



Coaxial feeding		Off-Axis feeding	
Pros	Cons	Pros	Cons
Higher flexibility	More complex nozzle design	Special nozzles are unnecessary	Less degrees of freedom
Symmetric deposition	More expensive head	Cheaper head	Non symmetric deposition

Coaxial: laser beam reach position, melt powder
→ head bring powder just on the place when it will be
↓
melt the necessary metal

(shield material to reduce air reaction)

↑
With COAXIAL it can be shield by the complex head

OFF-AXIS from nozzle the material exit and LASER melt
material on position → difficult shield and asymmetry
on deposition

Non symmetric

head →

feeding UNIT that create head moving obstacle

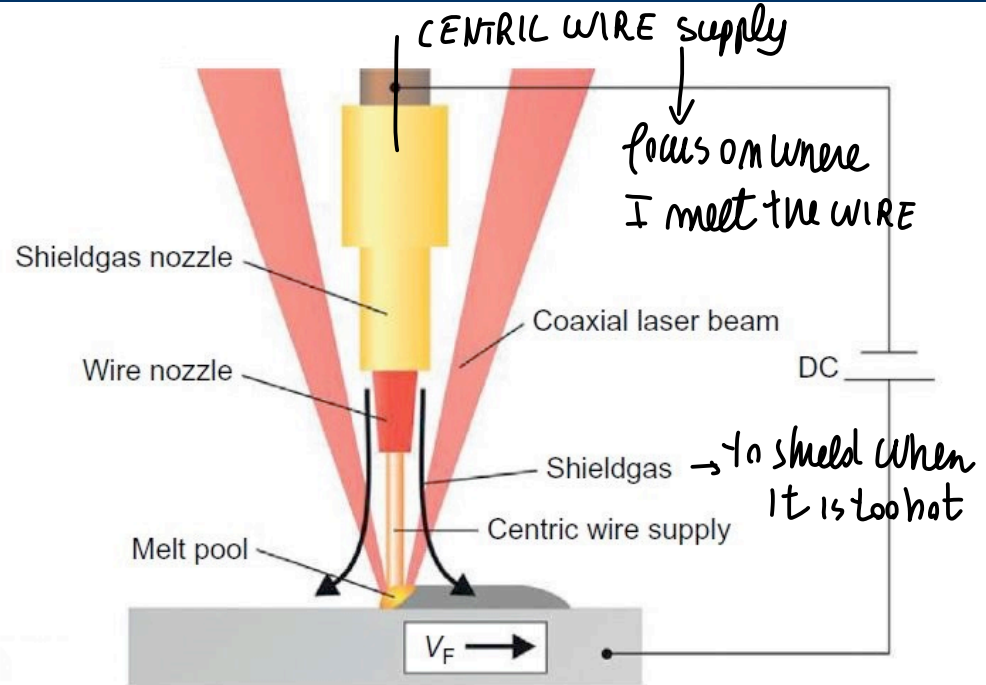
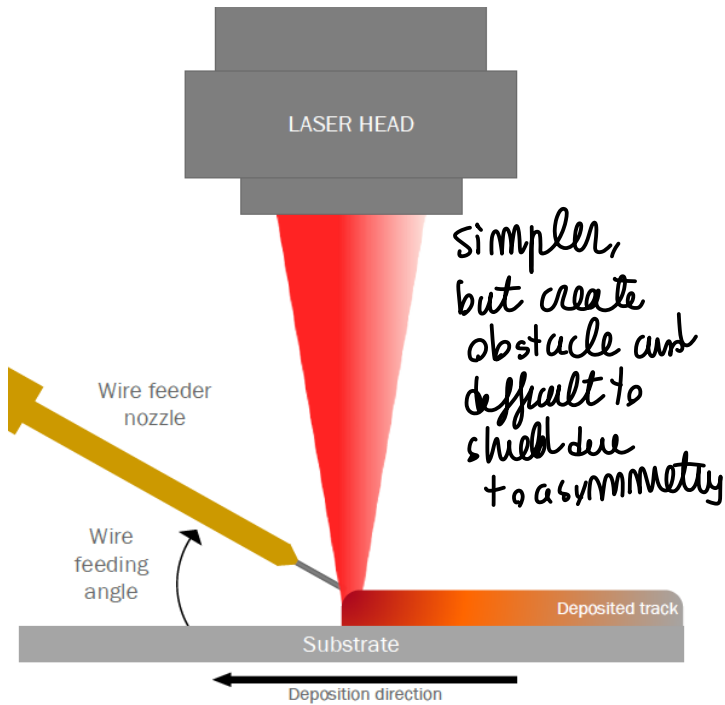
simple to obtain

but obstacle for geometry → low cost



LASER METAL DEPOSITION: WIRE FEEDING

→ same axial consideration



Off-Axis feeding		Coaxial feeding	
Pros	Cons	Pros	Cons
Cheaper Head	Non symmetric deposition	Symmetric deposition	Very complex nozzle design
Special nozzles are unnecessary	Less degrees of freedom	Higher flexibility	More expensive



POWDER BED FUSION VS DIRECTED ENERGY DEPOSITION

COMPARISON AMONG
PROCESSES

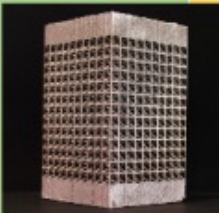
Assembly within build



High resolution



Powder
bed



Light-weight structures

Conformal cooling channels

Higher quality part, smaller here
just feed the material, lot of powder
so wasting cost sometimes powder is costly

using exactly the material that you need → feeding with the desired powder

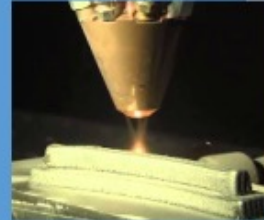
High deposition rate



Large parts



Directed
energy



Improved material use



Build on existing parts

Solution that try
to create layer part more rapidly
BIGGER OBJ
⇓



HYBRID PROCESS → using DED tech together machining

+ increase final accuracy of final object

2 KIND of tech

together in same machine → add + remove material
adding complexity to have better final qualities

SAUER

starty from DED powder depositing

MELT powder in position

Multistep → changing the tools

(removal processes) ← MACHINING
combined with AM

LOT of different
objects
accurate

⇒ YOU CAN OBTAIN BIG OBJECTS respect Metal Additive manufacturing



SOME NEW INTERESTING DIRECTIONS

Camozzi e Politecnico di Milano insieme per la ricerca e la crescita innovativa

↓ Under developing machine
(with multiple ISSUES)

EXTRUSION Based process to obtain BIG objects
for (electrical BOAT)
composite polymer based material

for manipulator
control problems

<https://www.youtube.com/watch?v=k0poVtBhlsQ>

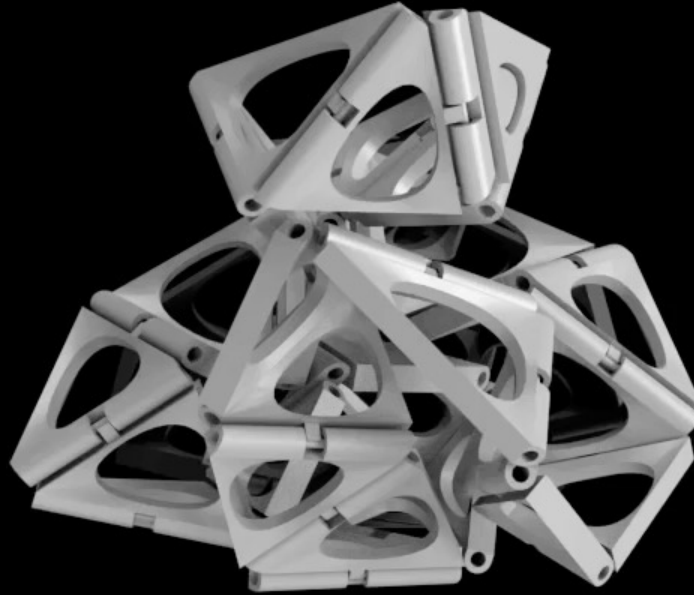


4D PRINTING

link selectively the components → possible evaluation,
CREATE a moving assembled
part → link components
with a GAP
movable

possibility to change
object shape
↓
NOT fixed geometry
[4D PRINTING]

depend on
element shape →
different behaviours
⇒ able to create
different elements



3D scanning
for designer
↓
simulation
(and complex
geometry)



SOME NEW INTERESTING DIRECTIONS

PASTA 3D Print → difficult shape of PASTA complex!

- max personalized
- extrusion based process



<http://www.3ders.org/articles/20160511-custom-made-pasta-italian-cuisine-giant-barilla-unveils-3d-pasta-printer.html>



3D PRINTING FOR A BETTER HUMAN LIFE

63

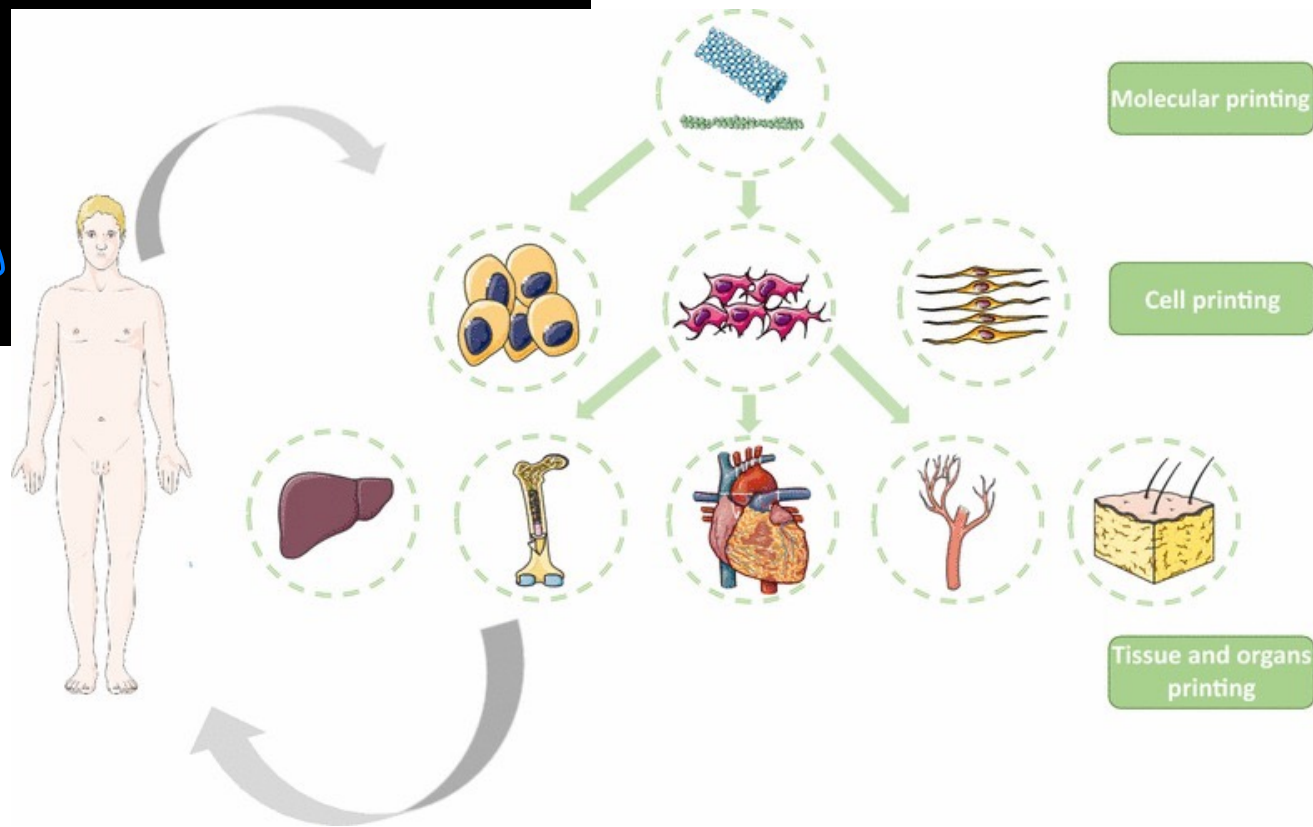
→ PRINTING for medicine use → BIOPRINTING tech using
complex tissue cells as glue
for medical resource ⇒ high STUDY
and DRUG development
|| disease study ||

↙ @ POLIMI
interlap studying
3D printing for
biological application
BIO-PRINTING

{ORGANOVO} {GOAL TO
create a
Body
organ !}

https://www.youtube.com/watch?time_continue=8&v=s3CiJ26YS_U

3D objects out
of cells
on going study





LUNAR APPLICATIONS

[ESA]

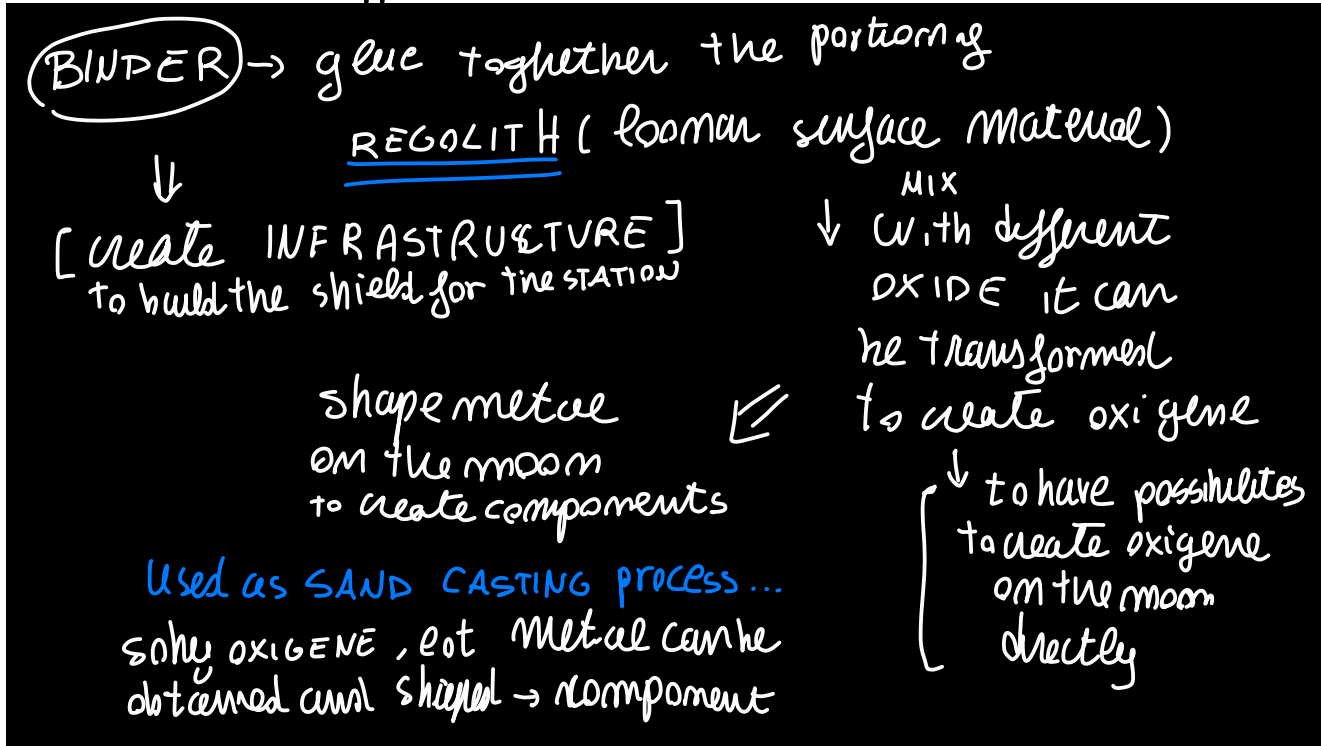
(Sustainable Building)

64

by launching the tech to the moon with a ROCKET
 ↳ robotic lander build the Humanbase before the arrival ⇒
 (3D Robot printer) + Drone support on LANDER

because moon
 ↳ NOT protected
 by Atmosphere by - Meteors
 · δ Radius
 · Temperature

3D PRINT
 ROBOT
 ↓
 Collect
 Regalite
 and PRINT
 Shield
 protection
 around the
 blow base



Usefull



ISSUE on creating parts → REGOLITE as sand casting MOLD material

{ AM for
 Moon building! }

Material Jet
 technologies to
 deal with metals (used with polymers)



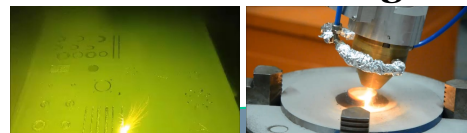
METAL ADDITIVE MANUFACTURING @ MECHANICAL ENGINEERING DEPARTMENT

65

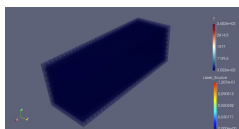
↓ POLIMI research ... create materials,
get INFO about process -- to make it BETTER → TEST



Manufacturing



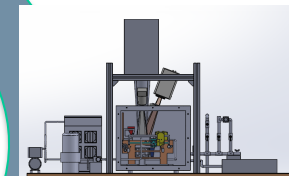
Simulation



Measurement



New solutions



Post-processing and finishing

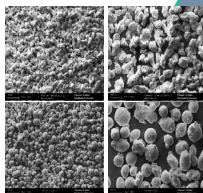
Qualification & testing

↑ INFO during the
process so you know
every reaction if it is Good

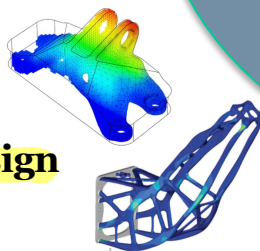
Challenges:

- Zero-defect AM
- New capabilities (*new materials, shapes, size, multi-material, functionally graded,...*)
- Increase productivity
- Reduce costs
- New solutions

Materials



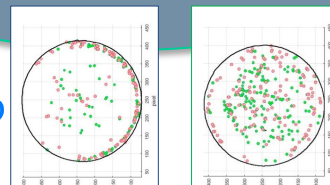
Design



Dealing with metals
and polymer based
objects...

EXTERNAL
control loop
ADAPTIVE
CONTROL

monitor defects during
the building phase





METAL ADDITIVE MANUFACTURING @ MECHANICAL ENGINEERING DEPARTMENT

66

End users

THALES Ansaldo STS



AddMe.Lab

Systems producers



Platforms/networks

Research institutions



Additive Manufacturing (5 CFU)
is among the courses of the TAB3 Group



prof. Bianca Maria Colosimo